









Seven strategies to prevent drowning

Technical package for policy-makers

World Health Organization

Seven strategies to prevent drowning: technical package for policy-makers

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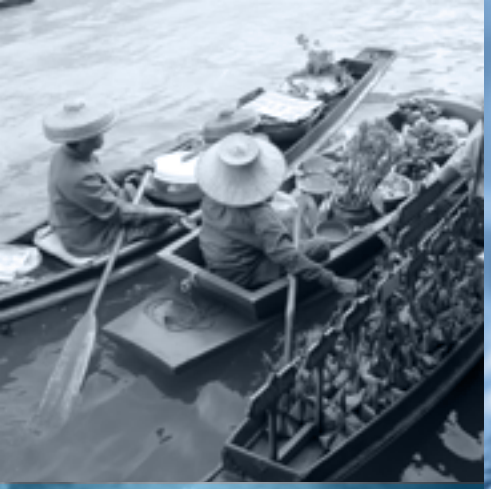
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Abbreviations

BFSCD	Bangladesh Fire Service and Civil Defence
CPR	cardiopulmonary resuscitation
ILCOR	International Liaison Committee on Resuscitation
ILO	International Labour Organization
IMO	International Maritime Organization
NGO	nongovernmental organization
PEER	Programme for Enhancement of Emergency Response
RNLI	Royal National Lifeboat Institution
SAR	search and rescue
SDG	Sustainable Development Goal
SOLAS	International Convention for the Safety of Life at Sea
UN	United Nations
WHO	World Health Organization





Executive summary and **introduction**

Young girl washes hands from uncovered water source,
© WHO / Yoshi Shimizu

Executive summary

Drowning is preventable, yet it remains a major global public health and development challenge that claims the lives of more than 300 000 people each year – nearly a quarter of these deaths are of children under the age of five years. Low- and middle-income countries carry the greatest burden, with risks rising due to climate change, extreme weather, migration, rapid urbanization, informal employment and reliance on passenger ferries and inland water transport.

The PROTECT technical package builds on earlier evidence-based WHO resources on drowning prevention. These set out why drowning is not an inevitable consequence of living, working, travelling or recreating in or near water, but rather a preventable cause of death that can be reduced through coordinated policy action.

PROTECT is therefore designed to provide national and local policy-makers working across sectors with a practical framework to accelerate collaborative national drowning prevention efforts through stronger laws, policies and systems across multiple sectors.

These can range from health, education, transport and maritime safety sectors, to occupational safety, disaster risk management, infrastructure, social welfare and child protection. PROTECT can also guide focal points working in individual sectors.

Seven PROTECT recommendations

Aligned with the Global Strategy for Drowning Prevention, PROTECT strengthens the drowning prevention policy environment by helping governments prioritize safety around water; focus on high-impact, evidence-informed actions to prevent drowning; and deliver effective community-level interventions. It is organized around seven key domains – adaptable to national contexts, priorities and capacities – and multiple ways to strengthen, expand or regulate them.

Each recommendation is supported by practical implementation guidance, illustrative case studies and a “pathway to PROTECT progress” that demonstrates how countries can strengthen drowning prevention systems over time.

These pathways recognize that countries are at different stages of development and provide flexible examples of how efforts can evolve from initial awareness and action towards integrated, sustainable and system-wide approaches.

Adopting a systems approach to drowning prevention

The PROTECT technical package offers a systems-oriented approach to drowning prevention, recognizing that effective prevention depends on coordinated action across multiple sectors rather than delivery through isolated activities.

A systems-oriented approach can address risk at its source, protect those most exposed and embed safety in the policies and institutions that shape everyday life. A four-step process outlined for each PROTECT recommendation can help achieve this – namely know the risks, make safety the standard, design for safety and keep it safe. A systems-oriented approach can lead countries from identifying risks and designing systems to implementing and sustaining interventions, while fostering cross-sector responsibility that long-term sustainability requires.

Generating co-benefits across sectors

As the seven PROTECT technical package recommendations are inherently interconnected, progress in one area can drive progress in others, generating multiple co-benefits that reinforce the overall drowning prevention system, simultaneously advancing multiple national priorities. Examples of co-benefits include improved child health and development, greater economic productivity, enhanced climate resilience, more inclusive access to recreation and water resources and reduced pressure on emergency and health services.

When seen this way, investment in drowning prevention becomes an investment in public health, community strengthening, sustainable development and social and economic resilience.

Conclusion

Implementing PROTECT requires political commitment, sustained investment and collaboration across sectors and levels of government, supported by communities, civil society and technical partners. Countries can apply the package in ways that reflect their own priorities, capacities and risk profiles, while building progressively towards stronger governance, better prevention and greater resilience.

By investing in drowning prevention, policy-makers can save lives, reduce inequities and advance broader goals for health, development and social well-being. The opportunity now is to translate this guidance into action, ensuring that safer interaction with water becomes a shared national responsibility and a lasting policy priority.

Image on next page:
Child securing life vest, © WHO / Petra Hongell



Physical infrastructure that promotes safe interaction with water.

Establish regulatory standards for safe public access at busy water settings.

Integrate drowning prevention into land-use planning, zoning regulations and infrastructure development policies.

Mandate protective barriers and fencing around busy aquatic settings.



Rescue and resuscitation capability for bystanders and professional responders.

Adopt a standard curriculum for safe rescue and resuscitation training, integrated into population-wide training and education programmes.

Mandate installation and regular maintenance of rescue and lifesaving equipment at high-risk or busy aquatic areas.

Institutionalize water-rescue capabilities in emergency response systems, following best-practice operational protocols.



Occupational safety for work around water.

Adopt international labour and maritime safety standards that protect workers exposed to drowning risks, developing sector-specific codes of practice.

Legislate for practical safety training, drills and certification for workers, tailored to the risks of different water-related occupations.

Mandate workplace safety systems for risk assessment, inspection, reporting and continuous improvement.



Transport safety standards for travel over water.

Align vessel safety standards with best-practice regulations, requiring routine inspection/certification of vessel structure, maintenance and lifesaving equipment.

Mandate training and certification for commercial vessel operators and crew in emergency preparedness, evacuation, rescue, first aid and communications.

Enforce systems to prevent overloading, manage passenger safety, restrict travel during hazardous weather and prevent alcohol/drug-impaired operation of vessels.



Education in basic swimming and water safety.

Institutionalize basic swimming and water safety education in national education, workforce development and community learning systems, using national standards.

Mandate instructor certification aligned with national standards and implement quality-assurance systems for training delivery.

Implement policies and finance that ensure equitable access to safe training environments, addressing location, gender, disability, age and financial barriers.



Child-care systems and safeguarding.

Institutionalize government-supported supervised child-care services within national child welfare and early childhood development systems.

Mandate safeguards and environmental safety standards to reduce young children's exposure to water hazards in homes, child-care settings and public spaces.

Require drowning prevention to be incorporated into caregiver education, parenting programmes and early childhood development frameworks.



Threat preparedness for floods and other multi-hazard emergencies.

Establish coordinated governance and institutional mandates for multi-hazard preparedness that define clear roles, responsibilities and standard operating procedures for early action, response and recovery.

Mandate development, operation and maintenance of national multi-hazard early-warning systems that provide timely, accessible and actionable risk information.

Establish financing mechanisms and operational frameworks that support community-level preparedness and emergency response capacity.



Introduction

Vision

Drowning is preventable. It can be reduced through policies that place safety at the heart of how people interact with water, creating enabling environments for evidence-based prevention measures and protecting those exposed to drowning risk through daily life, work, travel and recreation *(1)*. The PROTECT technical package helps countries do this by providing a framework to accelerate national drowning prevention efforts through stronger laws, policies and systems across multiple sectors, driven by collaboration.

Why drowning prevention is a multisectoral challenge

Drowning requires collaborative action across sectors including (among others) health, education, transport, maritime safety, occupational safety, disaster risk management, infrastructure, social welfare and child protection *(2)*.

The PROTECT technical package is therefore designed for policy-makers across many sectors, where decisions directly or indirectly contribute to establishing safer environments in general, including by reducing drowning risk.

Rationale

The global burden of drowning is high, making drowning prevention an urgent policy priority. Over 300,000 lives are lost to drowning each year – 24% of which are children under the age of 5 years. Low- and middle-income countries bear the heaviest losses (92%), with a drowning death rate more than three times higher than that of high-income countries *(1)*.

Many global threats fuel the already heavy drowning burden, from climate change and extreme weather events to sea-level rises and coastal inundation. Population shifts, migration, rapid and poorly planned urbanization, expansion of informal employment and use of passenger ferries and inland water transport all drive greater exposure to drowning risks, straining rescue, health and emergency services.

Over 300 000 lives are lost to drowning each year – 24% of which are children under the age of 5 years.

Why this package evolved

The 2021 UN Resolution 75/273 (2021) *(3)* and WHA resolution 76.18 (2023) *(4)* outline a series of evidence-based, cost-effective and scalable recommendations for drowning prevention and encourage Member States to implement them in ways that are appropriate to their national contexts. To support implementation of recommendations related to drowning prevention policy, legislation and regulation, WHO developed the PROTECT technical package to guide policy-makers.

The Global Strategy for Drowning Prevention *(5)*, developed by the WHO-hosted Global Alliance for Drowning Prevention and published in 2025, provides an overarching framework to drive coordinated action on drowning prevention from community to global levels (see Box 1).

The PROTECT technical package sets out practical and clear guidance on ways to do this. It also builds on several milestone resources published by WHO over the past decade or so, including the Global report on drowning prevention: preventing a leading killer (2014); Preventing drowning: an implementation guide (2016) *(6)*; Hidden depths: the global investment case for drowning prevention (2018) *(7)*; and the first Global status report on drowning prevention, published in 2024 *(1)*.

These resources, along with the WHO Guideline on the prevention of drowning through provision of day-care and basic swimming and water safety skills *(8)*, provide the technical, policy and funding guidance necessary to accelerate global action on drowning prevention.

Box 1. Alignment of PROTECT with the *Global strategy for drowning prevention*

The Global Strategy for Drowning Prevention *(5)* frames action across three intersecting domains – all people, all places and all contexts. PROTECT provides the policy and systems mechanisms needed to translate these domains into action.

All people - PROTECT prioritizes populations at greatest risk of drowning across the life course. Child-care systems and safeguarding measures help protect young children; swimming and water safety education build essential skills for children, adolescents and adults; and occupational safety measures support workers at heightened risk, including fishers and transport workers.

All places - PROTECT promotes safer environments wherever people interact with water. This includes strengthening physical infrastructure; improving safety within child-care settings; and establishing transport safety standards that protect people in homes, communities, workplaces, recreational settings and transport systems.

All contexts - PROTECT recognizes that drowning risk can arise during everyday activities, work, travel, recreation and emergencies. It addresses these risks through stronger rescue and resuscitation capability, improved occupational and transport safety and enhanced preparedness for floods and other multi-hazard events.

Together, the seven PROTECT domains provide a practical policy framework for implementing the *Global Strategy for Drowning Prevention* at country level. By organizing action across key policy areas, PROTECT helps governments translate the Strategy's priorities into legislation, regulation, systems and programmes that can be implemented across sectors and sustained over time.

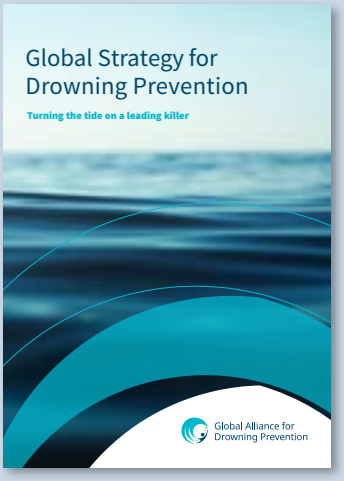
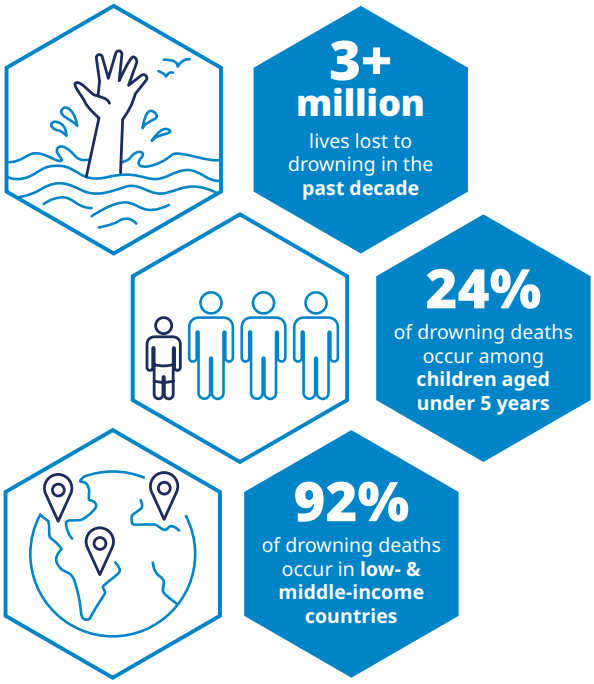


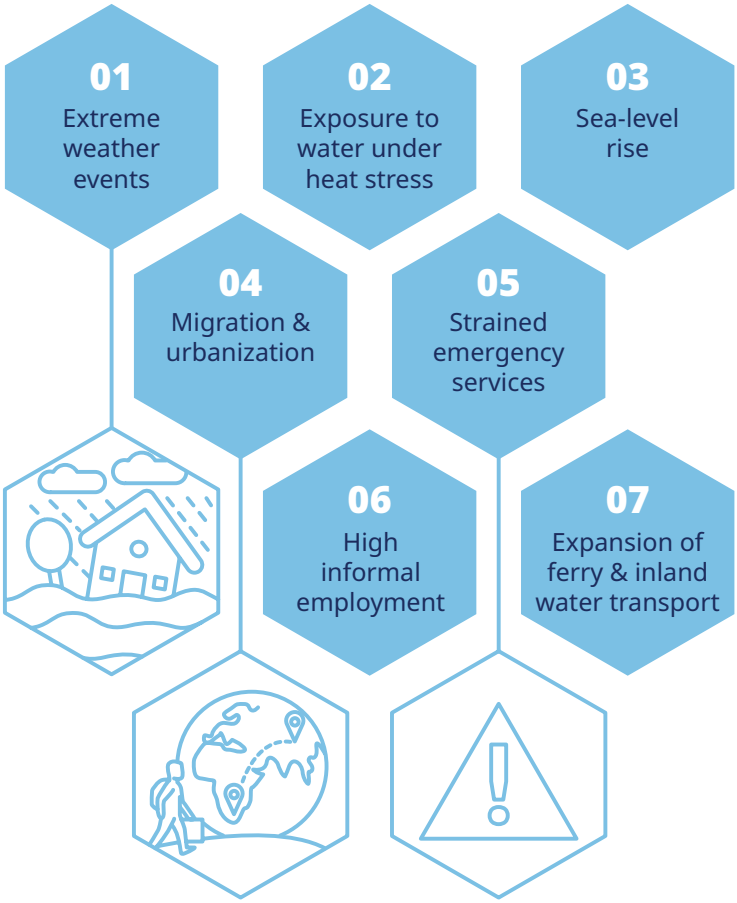
Fig. 1. Drowning prevention policy solutions and their expected national and global impacts

Current situation

Global burden of drowning



Emerging threats contributing to the drowning burden



Policy solutions

Recommended policy interventions for drowning prevention



EXPAND	ENHANCE	STRENGTHEN	REGULATE	EXPAND	REINFORCE	IMPROVE
Physical infrastructure that promotes safe interaction with water	**Rescue and resuscitation capability for bystanders and professional responders**	**Occupational safety for work around water**	**Transport safety standards for travel over water**	**Education in basic swimming and water safety**	**Child-care systems and safeguarding**	**Threat preparedness for floods and other multi-hazard emergencies**

Expected outcomes

Benefits of implementing recommended polices



Improved population health through reduction in preventable drowning deaths and injuries



Enhanced health and wellbeing



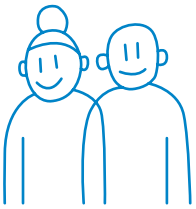
Strengthened child development



Increased lifelong learning opportunities



Enhanced economic productivity and protection of household livelihoods



Promotion of gender equality



Strengthened climate and disaster resilience



Reduction in future emergency response costs



Support of safe mobility, tourism and broader social participation

Global agendas progressed as a result

SDG	Sustainable Development Goals (specifically: 1, 2, 3, 4, 6, 8, 11, 13)
CRC	Convention of the Rights of the Child
UHC	Universal Health Coverage
ILO	International Labour Organization conventions
IMO	International Maritime Organization conventions
SENDAI	Sendai Framework for Disaster Risk Reduction
UNFCCC	United Nations Framework Convention on Climate Change and Paris Agreement

Purpose

The PROTECT technical package is designed to strengthen the policy environment for drowning prevention in three main ways:

- ensuring relevant national policies explicitly endorse safety around water, reflecting international guidelines, conventions and best-practice approaches;
- guiding prioritization of high-impact policy actions by spotlighting evidence-informed measures with the greatest potential to reduce drowning and identifying opportunities to strengthen implementation and enforcement;
- supporting and encouraging the delivery of evidence-informed drowning prevention interventions at community level.

While the main goal of this technical package is to reduce preventable drowning deaths, it is also designed to deliver co-benefits across multiple sectors and cross-cutting agendas, from health to economic productivity, and from strengthened climate and disaster resilience to supporting safer mobility and tourism.

Implementing mandates or laws can be challenging in some countries and sectors (e.g., informal or small-scale sectors), so this package offers flexible, evidence-based policy options that can adapt to each country's legislative, cultural and institutional realities. Policy-makers can use the package through a stepwise approach or refer to sections that are most relevant to them, whatever stage they are at in their drowning prevention journey.

The PROTECT technical package can also help strengthen countries' alignment with global commitments and frameworks for sustainable development (including multiple sustainable development goals, SDGs), universal health care and climate resilience, among others (Fig.1).

PROTECT: pathways to progress

The PROTECT technical package promotes a systems-oriented approach to drowning prevention that emphasizes coordinated, multisectoral action to reduce drowning risk and strengthen safety over time. Rather than relying on stand-alone interventions or placing primary responsibility on individuals, this approach recognizes that drowning risk is shaped by broader systems relating to health, education, transport, disaster preparedness, child care, labour, planning and community safety.

Across the seven PROTECT recommended actions, countries may strengthen drowning prevention systems in different ways and at different speeds depending on their context, priorities, capacities and existing structures. To help illustrate this, each recommendation includes a "PROTECT pathway" that provides examples of how efforts to prioritize water safety may evolve over time – from early awareness and basic safety actions, through stronger coordination and standards, toward more integrated and sustainable systems.

The pathways shown are intended as practical, flexible, illustrative examples of how to progress drowning prevention, rather than prescriptive drowning prevention stages or formal scoring framework (this role is met by the periodic *Global status report on drowning prevention*, dedicated to tracking progress on drowning prevention globally). Progress is unlikely to be linear and countries may be at different stages across different domains. The pathways therefore aim to support advocacy, dialogue and systems thinking by offering illustrative trajectories of potential milestones as drowning prevention evolves from being reactive towards a more coherent, mutually reinforcing and sustainable system.

Components

The PROTECT technical package contains three sections.

Section 1 offers a suite of best practice, evidence-based recommendations, adaptable to national contexts and burden. Each recommendation represents a distinct but interlinked policy domain, reflecting how governments can reduce communities' drowning risk and enable safe, sustainable and productive interaction with water through coordinated, multi-sectoral policy actions.

Section 2 contains overarching guidance on creating an enabling policy environment for implementing the package's seven recommended actions for reducing key drowning risks and sets out how the PROTECT package can be put to work through the key implementation steps, from positioning drowning prevention and building the case for it, to designing system-wide policies, promoting public engagement, and monitoring policy.

Section 3 outlines the rationale for applying a systems-oriented approach to the actions in this technical package.

Audience

The audience for this package comprises policy-makers and decision-makers in national and local government agencies with responsibility for developing, revising or overseeing the implementation of national policies, strategies and laws across sectors – from health, education, transport, maritime, and occupational safety, to disaster risk reduction, infrastructure, social welfare and child protection, among others.

The package is also designed to be practical and accessible for drowning prevention focal points working within individual sectors.

Methodology

This technical package was developed using a structured, evidence-informed process that reflects global best-practice and international development and maritime commitments; identifies and addresses evidence gaps; and supports a safe system approach to injury prevention. Drawing on a rigorous review of existing evidence, this process combined expert consensus and practitioner insights from diverse country contexts.

A multidisciplinary group guided the development of the package from the outset (including technical experts in drowning and injury prevention and in public health, policy, and programme implementation), validating methods and interpretation of findings at every stage. Each working group member signed a Declaration of Interests to protect the integrity, credibility and independence of the project.

A systematic review of literature identified effective policies and implementation approaches for drowning prevention in many contexts and settings, focusing on population-level impact, feasibility and equity *(9)*. These findings were used to identify key policy domains for the PROTECT technical package and to create an initial framework of recommended actions based on the strongest evidence.

Where published evidence was lacking, a Delphi study and in-depth interviews with technical specialists helped establish priority actions and best practices, ensuring recommendations were appropriate, robust and informed by real-world experience.

Recommendations for each domain were organized using an adapted Exploration, Preparation, Implementation, Sustainment (EPIS) Framework *(10)* – an established implementation science model. Mapped to the EPIS framework, four dimensions were specifically developed to reflect a systems-oriented approach to drowning prevention through the PROTECT technical package (know the risks, make safety the standard, design for safety and keep it safe). Together they guide countries from identifying risks and designing systems to implementing and sustaining interventions, while fostering cross-sector responsibility that long-term sustainability requires.

The project working group reviewed draft recommendations for coherence, technical accuracy and consistency with WHO guidance and systems-oriented principles. Their feedback enhanced clarity and applicability, resulting in a practical, evidence-informed set of policy actions for accelerating policy progress on drowning prevention.



Image on next page:
Passengers travelling by small boats across a river, © WHO / FCTC / Vlad Sokhin







Section 1

PROTECT recommendations

Children play at base of waterfall,
© WHO / Deanna Gerlach

PROTECT recommendations

This section sets out the seven PROTECT policy recommendations for drowning prevention (Fig. 2). Although each recommendation can be implemented independently, the greatest impact is achieved when they are combined into an overarching, multisectoral, national drowning prevention strategy. Such an approach helps align efforts across ministries and agencies, creates coherence between national and subnational levels, leverages available resources and expertise and strengthens sustainability.

By embedding drowning prevention within broader policy agendas – such as child health, occupational safety, recreation, tourism, climate resilience and disaster risk reduction – countries can ensure that actions are integrated, scalable and sustained.

Fig. 2. PROTECT recommended actions



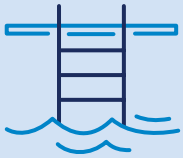
P	R	O	T	E	C	T
EXPAND	ENHANCE	STRENGTHEN	REGULATE	EXPAND	REINFORCE	IMPROVE
Physical infrastructure that promotes safe interaction with water	**Rescue and resuscitation capability for bystanders and professional responders**	**Occupational safety for work around water**	**Transport safety standards for travel over water**	**Education in basic swimming and water safety**	**Child-care systems and safeguarding**	**Threat preparedness for floods and other multi-hazard emergencies**

Image on next page:
Two children stand at seaside,
© WHO / Karen Reidy

By embedding drowning prevention within broader policy agendas countries can ensure that actions are integrated, scalable and sustained.



P



Expand

Physical infrastructure that promotes safe interaction with water

Develop infrastructure to support safe interaction with water (and prevent unsafe access) in public, private and household settings, informed by proven standards and underpinned by local enforcement.

What to do?

- **Establish** regulatory standards to enable safe public access to water at busy aquatic settings, including requirements for designated safe entry and exit points, signage and rescue equipment.
- **Integrate** drowning prevention into land-use planning, zoning regulations and infrastructure development policies.
- **Mandate** protective barriers and fencing around busy aquatic settings, including public and private swimming pools and easily accessible water bodies frequented by children, particularly around the home.

Why do it?

Engaging with water is often essential for daily life, be it collecting water from ponds or rivers for washing, cooking, or bathing; travelling or working on waterways or seas; or visiting aquatic locations for health, well-being and recreation.

Water also provides opportunities to learn vital swimming and safety skills that can save lives. Whatever the context or setting, safe access to water should be promoted and ensured for all.

Well-designed water safety infrastructure can reduce drowning risk and support health, climate resilience and sustainable development, including achieving multiple international development targets (see Box 2).

Box 2. Related international commitments

- SDG 6.1: Achieve universal access to safe water
- SDG 11.2 and 11.7: Provide safe, accessible public spaces
- WHO/UNICEF Early Childhood Development Framework: Create safe and enabling environments for young children as part of holistic early childhood development

Co-benefits for communities and multiple sectors can include:

- building public confidence and a culture of safety, encouraging people to behave responsibly around water and to act quickly in emergencies;
- reducing loss of life and long-term disability, lowering health and social care costs associated with drowning and other water-related injuries;
- strengthening links between community responders, professional response services and health care providers for more efficient care; creating employment and volunteer opportunities through certified lifesaving and first-aid training networks;
- a network of trained first responders that can respond to illnesses or injuries beyond drowning;
- greater population resilience to water-related emergencies, lowering fatality risks during climate events and reducing risk of multiple fatalities during rescue attempts;
- improved safety and public confidence around water, supporting tourism, transport, fisheries and other water-based economic activities.

How to do it

Know the risks

- Conduct risk assessments to identify specific populations, locations and water-access points for which interventions are required, taking account of seasonal and use-related risks.
- Consult communities to understand local behaviours, preferences, cultural norms and perceived risks, and ensure interventions are appropriate, acceptable and sustainable.

Make safety the standard

- Adopt and enforce policies aligned with best practice legislative standards where available, such as swimming pool fencing requirements (see Box 3).
- Integrate safety standards into zoning codes and land use planning, including:
  - establishing safe zones to ensure safe recreation in and around water;
  - requiring construction to include protective measures when building near high-risk waterbodies;
  - requiring protective infrastructure (such as fencing) around high-risk water bodies;
  - incorporating flood risk reduction into land-use planning.
- Apply universal design principles[i] when planning infrastructure to ensure safe and equitable access for all, including children, older people and people with disabilities. Apply the principle of “easy to get into, easy to get out of the water”.
- Promote the use of simple, low-cost, childproof barriers, such as fencing, gates, covers and enclosures, to prevent young children from accessing nearby water hazards around the home.

i Universal design principles aim to create products, environments and systems that are usable by all people, to the greatest extent possible, without the need for adaptation or specialized design.

How to do it

Design for safety

- Design busy water-access locations to ensure clear visibility of users and safe and rapid rescue access and facilitate routine and practical maintenance.
- Build entry and exit points at low-risk or busy water-access areas using non-slip surfaces, gentle slopes and handrails. Where possible, ensure no sudden increase in depth.
- Clearly designate safe entry and exit points into water bodies that protect workers and support safe and efficient operations.
- Use landscaping (such as vegetation or terracing) to guide users to safe access points and away from hazards.
- Establish designated entry/exit points with lifesaving equipment.
- Install low-glare lighting on routes used at dawn and dusk to help people see water.
- Keep safety equipment, signage and rescue equipment visible at high-use locations.
- Clearly mark changes in water depths or currents, where necessary.
- Where feasible, install barriers to prevent access to easily accessible water bodies that are in locations frequented by children and reinforce the need for adult supervision.
- Design barriers such as rails and chains so they cannot trap anyone, especially children.

Box 3. Swimming pool fencing requirements

Fences around public and domestic swimming pools should be four-sided, at least 1.2 metres high, constructed from durable, non-climbable materials and include self-closing, self-latching gates opening away from the pool, while ensuring no direct access from dwellings unless protected by alarms or child-resistant, self-closing doors *(11)*.

Box 4. Monitoring and enforcing proper infrastructure use and maintenance

Effective water safety infrastructure depends on routine inspection, maintenance and enforcement, alongside good design. Local governments can establish inspection programmes for aquatic facilities and high-risk water sites, supported by clear standards, reporting systems, licensing processes and accountability mechanisms. Integrating water safety checks into broader public health, urban planning, and disaster risk reduction systems can improve sustainability and compliance *(12)*.

Keep it safe

- Regularly inspect and repair barriers, handrails, steps, decking and other infrastructure, especially after storms or flooding.
- Monitor and enforce proper use of infrastructure, including inspection programmes led by local authorities (see Box 4).
- Adapt site design to meet changing community use, needs and local conditions.
- Maintain community involvement to ensure use, local ownership and lasting upkeep.

Key partners and sectors to involve

Preventing drowning through physical infrastructure involves many sectors and actors:

- **Government** sectors include housing and infrastructure, urban planning/design/management, environmental and natural resources agencies, tourism, health and sports and recreation.
- **Nongovernment** actors include community organizations and local user groups, landscape architects, land-use planners, engineers and builders, standards and certification agencies and outdoor recreation associations.

PROTECT pathway to progress

A "systems maturity pathway to progress" for this recommendation is set out in Fig. 3. The pathway is neither prescriptive nor linear but aims to show how expanding physical infrastructure to enable safe interaction with water can evolve over time to embed safety as a core value of all related drowning prevention policies and interventions.

Fig. 3. PROTECT pathway for the recommendation "Expand Physical infrastructure that promotes safe interaction with water"



Case study 1

Revitalizing urban waterfronts to promote safe recreation around water, Estonia

This case study examines the health, safety and well-being impacts of structural improvements to a popular swimming spot in Kopilranna, Estonia.

The Estonian district of Kopilranna, in the capital Tallinn, has excellent sea views and is used all year round by residents for recreation around the water – including swimming, fishing, sunbathing on nearby rocks, picnicking, dog walking and ice-skating.

Kopilranna's popularity as an all-season water spot grew because it is the only place in the area with access to deeper water – despite the entry point traditionally being a very shallow concrete slipway that was not suitable for swimmers.

In 2019, the Estonian University of Life Sciences, Exeter University and the BlueHealth project collaborated to co-design a community space to meet local needs in Kopilranna and promote health and well-being activities through connectedness to nature.

Several design options were developed in consultation with local people. This process included meetings and workshops to identify existing uses and possible conflicts of use in relation to the site; questionnaires for locals delivered during local festivals; and a public vote on the final three design options.

While the intervention did not alter the old slipway itself (for legal reasons), a sundeck with steps was constructed alongside it, with handrails, and more seating to enable parents and carers to supervise children playing near the water, reducing drowning risk.

Overall, the new design shifted the area's derelict appearance, making it appear safer and more valued while promoting the psychological and social benefits of being close to the water. More people now use the site, including more women and children, which is an indirect sign of perceived safety in the area.

Image: Redeveloped waterfront with improved public access to water, © BlueHealth



Case study 2

Community-pond fencing protects young children in West Bengal, India



This case study demonstrates how low-cost, locally adapted infrastructure interventions, supported by community engagement, can effectively reduce drowning risk among young children in high-exposure rural settings.

Most drownings among children aged 1–4 years occur within 100 metres of home, especially in rural areas near open water, where caregivers often face multiple competing household and livelihood responsibilities alongside full-time child care. The George Institute for Global Health, supported by the Indian Council of Medical Research and the Child in Need Institute, launched a one-year programme for 100 households in the Sundarbans region of West Bengal, India, helping install bamboo and fishing-net fences around ponds near the homes of young children to prevent drowning.

The project helped the community identify households with young children that were located near open water; provided locally sourced bamboo and fishing nets; guided construction of fencing (including construction of self-locking gates); and monitored fencing maintenance throughout the year. Acceptance was high, with 30 additional families beyond the projects' scope motivated to build their own fences.

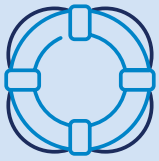
Families were more engaged following discussions about recent drowning events linked to ponds in communities.

Regular community meetings helped participants with fencing upkeep, though maintenance was harder for ponds in mangrove areas where ground conditions were challenging. The programme found that low-income or women-led households were more likely to need additional support with labour or obtaining materials and that timing community engagement for when men were available was important, as their buy-in was often critical for fence construction.

An evaluation found that by the end of the year, 95% of fences had their poles still securely attached and 97% had panels in place and functioning locks. All participants were satisfied and believed the programme to be effective in preventing child drowning deaths. No fatal or nonfatal drowning events were reported in the fenced ponds.

Image: Community-installed self-latching gate around pond, © George Institute for Global Health

R



Enhance

Rescue and resuscitation capability for bystanders and professional responders

Establish policies and systems that equip bystanders and professionals with the skills, training and resources needed to respond effectively to drowning incidents and water related emergencies.

What to do?

- **Adopt** a standardized curriculum for safe rescue and resuscitation training and integrate into population-wide training and education programmes.
- **Mandate** installation and regular maintenance of rescue and lifesaving equipment at high-risk or busy aquatic areas.
- **Institutionalize** water-rescue capabilities in emergency response systems, following best-practice operational protocols.

Why do it?

A population trained in safe rescue and resuscitation can provide a rapid response and help save lives. Early intervention can prevent a drowning event from becoming more serious or life threatening *(13)*.

Building both bystander and professional rescue and resuscitation capability is essential. Bystanders need the knowledge, skills and resources to respond to drowning incidents safely, effectively and without delay – protecting both themselves and the person they are trying to rescue *(14)*.

Professional emergency response systems (national or local) need clear leadership, coordination and adherence to recognized standards of practice to ensure an effective response to large-scale or complex events *(15)*. Strengthening capacity in rescue and resuscitation not only reduces deaths and injuries, it also increases disaster resilience, in line with international commitments (see Box 5).

Box 5. Relevant international commitments

- Sendai Framework for Disaster Risk Reduction – Priority 4: Enhance disaster preparedness for effective response
- SDG 3.2 and 3.4: Reduce preventable deaths and injuries
- Universal Health Coverage (UHC) Mandate: Framing bystander rescue and cardiopulmonary resuscitation (CPR) skills as a universal public health right aligns with SDG 3 and UHC mandates
- International Convention on Maritime Search and Rescue (SAR), 1979: Provide adequate SAR services and establish common procedures and training, among others *(16)*

Co-benefits for communities and multiple sectors can include:

- building public confidence and a culture of safety, encouraging people to behave responsibly around water and to act quickly in emergencies;
- reducing loss of life and long-term disability, lowering health and social care costs associated with drowning and other water-related injuries;
- strengthening links between community responders, professional response services and health care providers for more efficient care; creating employment and volunteer opportunities through certified lifesaving and first-aid training networks;
- a network of trained first responders that can respond to illnesses or injuries beyond drowning;
- greater population resilience to water-related emergencies, lowering fatality risks during climate events and reducing risk of multiple fatalities during rescue attempts;
- improved safety and public confidence around water, supporting tourism, transport, fisheries and other water-based economic activities.

How to do it

Know the risks

- Use drowning and rescue data to identify high-risk locations, scenarios and populations.
- Map current (professional and bystander) rescue and resuscitation capacity and availability of training and equipment.
- Analyse incident reviews and post-rescue debriefing sessions to identify system failures and prevention opportunities, treating each rescue as a chance to learn.

Make safety the standard

- Develop a standardized safe rescue and resuscitation training programme and identify strategies to train and retrain community members (see Box 6).
- Mandate rescue and resuscitation training for groups who frequently interact with water (e.g. fishers, boat operators, maritime workers, marine tourism providers), tailored to their skills, roles and level of exposure.
- Strengthen legal protections for bystanders and first responders who voluntarily help in water-related emergencies.
- Ensure rescue and resuscitation policies are responsive to cultural and social contexts, including faith-based considerations and gender dynamics, particularly in situations where cultural norms may affect bystander response or willingness to intervene.
- Develop effective systems for training, certification, equipment maintenance and post-incident data collection.
- Establish national coordination mechanisms to ensure all rescue services (e.g. marine, flood, coastal and beach rescue) operate through aligned systems, shared communication protocols and coordinated ways of working.
- Appoint a lead professional rescue agency (or multiple agencies, as this may differ between maritime and inland environments) for rescue operations at national or local levels.

How to do it

Design for safety

- Establish accessible rescue stations at busy, high-risk aquatic locations, equipped with appropriate lifesaving devices and clear guidance for bystander response (see Fig. 4 as an example).
- Ensure professional rescue teams follow recognized best-practice protocols, supported by regular training, drills and operational-readiness checks (guided by appointed lead agency) (Box 7).
- Where no professional search and rescue service exists, build the capacity of relevant government or nongovernmental services (e.g. fire services) to institutionalize water rescue capabilities in their emergency response systems (see case study: Bangladesh).



Fig. 4. Rescue station in Chiang Mai, Thailand

Box 6. Safe rescue and resuscitation training standards

A structured, written curriculum helps align training as closely as possible with relevant international guidance, including the International Liaison Committee on Resuscitation (ILCOR) treatment recommendations) *(15)*. International guidance, including from the International Life Saving Federation, emphasizes non-contact "reach and throw" rescue techniques using aids with water entry taught only as a last resort, minimizing risk to the rescuer *(17)*.

Box 7. Search and rescue best-practice protocols

Search and rescue best-practice protocols emphasize rapid activation, clear command structures and coordinated multi-agency response, in line with internationally recognized standards such as those promoted by the International Maritime Organization (IMO) (see https://www.imo.org/en/ourwork/safety/pages/imo-documents-relevant-to-sar.aspx) *(18)* and the International Maritime Rescue Federation (see https://www.international-maritime-rescue.org/listing/category/manuals-reports-and-guidance) *(19)*. Evidence-based guidance highlights the importance of trained responders, interoperable communication systems and regular simulation exercises to ensure effective, safe and timely rescue operations.

Keep it safe

- Maintain continuous joint training exercises and multi agency training events, re-certification and equipment inspection programmes to ensure readiness and reliability.
- Monitor rescue incidents and response outcomes to identify patterns, gaps and opportunities for prevention.
- Engage emergency services in proactively identifying hazards and educating the public to prevent drowning incidents.

Key partners and sectors to involve

Preventing drowning through safe rescue and resuscitation training involves many sectors and actors:

- **Government** sectors include health, disaster risk reduction, education, police, coastguard, fire and civil defence.
- **Nongovernment** actors include lifeguards, sports organizations, swimming teachers, search and rescue organizations, commercial rescue providers, tourism operators, technology firms, local responders, volunteer groups and community based organizations.

PROTECT pathway to progress

The pathway to progress for this recommendation is set out in Fig. 5. The pathway is neither prescriptive nor linear but aims to show how enhancing rescue and resuscitation capacity for bystanders and professional response systems may evolve over time to sustainably equip lay people and professionals with the skills, training and resources to respond safely and effectively to drowning incidents.

Fig 5. PROTECT pathway for the "Enhance Rescue and resuscitation capacity for bystanders and professional response systems" recommendation



Case study 3

Embedding flood rescue capacity in Bangladesh's National Fire Service

This case study illustrates the power of policy to generate systems change. Embedding flood rescue into an existing national agency shows how policy and institutional reform can deliver scalable, sustained reductions in drowning risk.

Formally appointing a national body with the appropriate expertise, training and equipment helps ensure flood rescue operations are coordinated, safe and effective during emergencies. Before 2013, the Bangladesh Fire Service and Civil Defence (BFSCD) held the country's national mandate for flood response but had no formal water-rescue training or in-built flood rescue capability.

In a country highly exposed to seasonal flooding and climate-related hazards, this created a systemic gap that placed many at risk, especially children, people with limited mobility, households in isolated areas and communities in remote rural districts. To address this gap, the Royal National Lifeboat Institution (RNLI) and BFSCD launched a Flood Rescue Project in 2013, aimed at building institutional flood rescue capacity within the national emergency service.

The initiative focused on system-level reforms, including integrating flood rescue training into BFSCD's institutional structure.

A flood rescue module was incorporated into standard firefighter training, while national training manuals were developed and training-of-trainers programmes were introduced to build in-country instructor capacity. Equipment such as personal flotation devices and throw-bags were locally manufactured to reduce long-term costs, while training materials were translated into Bengali to strengthen national ownership. The programme was deliberately designed as a low-cost, scalable model and was expanded nationally between 2016 and 2019.

By 2025, more than 500 responders had been trained across 14 districts, strengthening the country's ability to respond to major flood events. BFSCD can now mobilize rapid-action rescue teams within 12 hours during a major flood, enabling evacuation of vulnerable populations and coordinated search operations. Beyond operational improvements, the programme has strengthened interagency coordination, increased community trust in emergency services and enabled regional knowledge-sharing with neighbouring countries. The experience demonstrates how embedding flood rescue capacity in a national emergency service can provide a sustainable and scalable policy approach to reducing drowning risk in flood-prone settings while supporting broader disaster risk reduction and climate-resilience goals.



Case study 4

Strengthening rescue readiness along urban waterways, United Kingdom

This case study demonstrates how targeted policy-supported training and partnership-based delivery can strengthen early rescue capacity in high-risk environments linked to alcohol consumption and urban waterways.



Many drownings in the United Kingdom happen near pubs, bars and restaurants by the water, where alcohol can affect people's judgement and awareness around rivers, canals, harbours and marinas.

Venue staff and security personnel are often the first to witness someone falling or entering the water, yet may lack training in hazard awareness, safe-rescue techniques, or access to basic bystander rescue equipment. As emergency services may not always arrive quickly, early bystander intervention can be critical for saving lives.

Early efforts to address this issue began around 2010 at Teddington Lifeboat Station in London, located on a busy stretch of the River Thames with many pubs, restaurants and bars. Lifeboat crews delivered informal awareness sessions at local venues, setting out the potential benefits of equipping venue staff with basic rescue knowledge. As the concept developed, the Royal Life Saving Society UK helped clarify the policy and a legal framework for training provision.

A distinction was made between accredited water safety training that may be required through workplace risk assessments and the RNLI Waterside Responder Scheme, which provides voluntary, community-based training. This approach helped ensure legal clarity, safety and scalability while avoiding regulatory burdens that might discourage venue participation.

Building on the early pilot, RNLI developed a standardized training programme *(20)* covering hazard awareness, prevention messaging and safe throwline use. A pilot in 2016–2017 refined instructor training and identified practical challenges related to venue engagement and delivery logistics. RNLI's research indicated that training 100 venues could prevent approximately one drowning per year, supporting the case for scale up.

Governance of the programme remains with RNLI, but delivery typically occurs through partnerships with local agencies – most often fire and rescue services working alongside local water safety forums. The Tyne & Wear Fire and Rescue Service has helped roll out the programme to local communities along the River Tyne, training over 300 people in 18 waterside businesses. To date, there have been 14 confirmed incidents where individuals trained through the programme successfully intervened to save a life.

By clarifying legal responsibilities, removing cost barriers for participating venues and embedding delivery within existing community safety partnerships, the model provides a scalable approach that can complement formal rescue services.

Image above: *Throw bag rescue training along an urban waterway, © RNLI*
Opposite: *Emergency responders equipped for water rescue operations, © BFSCD*

O



Strengthen

Occupational safety for work around water

Adopt and enforce policies aligned with best-practice labour standards to protect the lives and livelihoods of those working in or near water.

What to do?

- **Adopt** international labour and maritime safety standards that protect workers exposed to drowning risks, developing sector-specific codes of practice.
- **Legislate** for practical safety training, drills and certification for workers, tailored to the risks of different water-related occupations.
- **Mandate** workplace safety systems for risk assessment, inspection, reporting and continuous improvement.

Why do it?

Many occupations require close and frequent contact with water, from fishing, aquaculture, farming and transport, to construction, sanitation and irrigation. Financial or employer pressure can push the (often informal) workers in these occupations to operate in hazardous conditions, without adequate training, protection, or access to safety equipment.

Ensuring worker safety on or near water (drawing on guidance from IMO, the International Labour Organization (ILO) and the Food and Agriculture Organization (FAO)) helps protect lives and livelihoods, strengthens communities, boosts productivity and advances sustainable development, in line with multiple international commitments (see Box 8). With growing interest in the blue economy[ii] and efforts to sustainably manage and conserve the oceans, ensuring water safety for those working around the water can also be applied to those involved in conservation and related work.

Co-benefits for communities and multiple sectors can include:

- fewer work-related injuries and deaths, lessening their human, social and economic impact on individuals, families and communities;
- greater sustainability through improved worker health, more resilient economies and socially and environmentally responsible practices;
- enhanced productivity through a healthier, better protected and more motivated workforce;
- reduced financial losses associated with workplace injuries and fatalities, including lower compensation, insurance and emergency response costs for governments, employers and insurers;
- stronger long-term viability for key economic sectors such as fisheries, agriculture, transport and construction.

Box 8. Related international commitments

- SDG 8.8: Protect labour rights and promote safe and secure working environments for all workers, including those in precarious employment
- ILO Conventions: Occupational Safety and Health (No. 155 and No. 188), Work in Fishing Convention (No. 188), Dock Work Convention (No. 137), Maritime Labour Convention (No. 186), Safety and Health in Construction (No. 167)
- IMO Conventions: International Convention for the Safety of Life at Sea (SOLAS) Convention, Cape Town Agreement, International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, International Convention on Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel
- Voluntary FAO, IMO and ILO instruments including Code of Safety for Fishermen and Fishing Vessels (2005), Voluntary Guidelines for the Design, Construction and Equipment of Small Fishing Vessels (2005), Document for guidance on training and certification of fishing vessel personnel (2001), Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries (2015)

How to do it

Know the risks

- Identify occupations, tasks and workplaces with high exposure to drowning risk, including in informal, seasonal and artisanal contexts, using workplace observation and consultation with workers and supervisors (see Box 9).
- Conduct a structured and risk assessment of the setting, identifying water hazards, workers most exposed to drowning risk, existing water safety practices and practical opportunities for strengthening them. In informal or low-resource settings, use guidelines and approaches that are appropriate to the local context.
- Review existing drowning prevention safety practices, regulations and enforcement mechanisms to identify gaps in protection, compliance and capacity to prevent water-related occupational injuries and drowning.

Box 9. Performing workplace risk assessments

The ILO provides practical guidance for improving occupational safety and health through participatory workplace assessments that identify hazards, engage workers and supervisors and prioritize feasible safety improvements. This approach is particularly useful in small enterprises and informal workplaces where drowning risks are often highest. A useful resource is the ILO's Training package on workplace risk assessment and management for small and medium-sized enterprises *(21)*, which includes tips for how to engage workers directly in the assessment.

ii The blue economy refers to the sustainable use of ocean, coastal and inland water resources to support economic growth, livelihoods and jobs while maintaining ecosystem health.

How to do it

Make safety the standard

- Adopt international conventions on worker safety and labour standards and align national laws; develop related codes of practice and safety guidelines to address occupational drowning.
- Strengthen the capacity of national institutions responsible for training and oversight to ensure effective local implementation of national policies at community level – e.g. by developing training materials in local languages and providing regular, practical safety training, certification and drills for workers in water-related occupations.
- Establish clear roles, responsibilities and coordination mechanisms among government agencies, employers, workers' organizations and industry groups to strengthen implementation and oversight of workplace water safety measures.
- Support action to ensure access to affordable and appropriate quality buoyancy devices.
- If legal change is slow, help foster a safety culture and support local, evidence-based drowning prevention solutions that can pave the way for later adoption of occupational safety and health standards.

Design for safety

- Provide safety training (including in use of protective gear such as personal flotation devices, rescue equipment and communication devices), certification and drills for water-based workers, tailored to their roles and exposure.
- Use posters, radio and community events to embed a safety culture in workplaces and surrounding communities.
- Co-design or create local safety protocols including response systems (e.g. local emergency response plans) to support local community ownership.
- Establish health and safety enforcement mechanisms, including risk assessments and hazard management, using licensing, inspection and reporting, supported by skilled public officials.
- Support workplaces to implement occupational health and safety management systems that enable routine monitoring and compliance (Box 10).

Box 10. Establishing workplace occupational health and safety systems

Supporting workplaces to implement occupational health and safety management systems requires clear frameworks, practical guidance and ongoing oversight. Governments can promote simple risk-management approaches to build the capacity of employers and workers. Routine monitoring through inspections and reporting and tracking of key indicators supports accountability, while coordinated action and enforcement help drive compliance and continuous improvement (see the ILO's Guidelines on implementing occupational safety and health management systems) *(22)*.

Keep it safe

- Conduct ongoing inspections through the health and safety enforcement mechanisms (included at design stage), refresher training and drills to maintain safety standards and readiness.
- Monitor compliance and incident data to assess effectiveness and emerging risks, using findings to refine guidance and enforcement.
- Strengthen reporting mechanisms to promote accountability among employers and across sectors.
- Use lessons learned to expand and strengthen occupational safety standards, reducing drowning risk while improving overall worker protection.

Key partners and sectors to involve

Preventing drowning through strengthening occupational safety around water involves many sectors and actors:

- **Government** sectors include labour, transport, capture fisheries and aquaculture, agriculture, infrastructure and police.
- **Nongovernment** actors include worker cooperatives and trade unions, employer associations, private industry, fisher associations and insurance providers.

PROTECT pathway to progress

The pathway to progress for this recommendation is set out in Fig. 6. The pathway is neither prescriptive nor linear but aims to show how strengthening occupational safety for work around water may evolve over time to sustainably to teach, legislate for and track worker safety to reduce their drowning risk.

Fig 6. PROTECT pathway for the "Strengthen Occupational safety for work around water" recommendation



Case study 5

Safe Fishing: Strengthening occupational safety culture in the fishing fleet, Norway





This case study illustrates how occupational safety regulations can be strengthened through system-wide collaboration between authorities, industry organizations and safety actors.

Commercial fishing is one of Norway's most hazardous occupations. Many deaths happen when fishers fall overboard or when small vessels capsize or founder. Risk factors include fishing alone, incorrect use of personal flotation devices, lack of personal safety equipment, normalization of risk in everyday work, harsh environmental conditions and time pressure while fishing.

Fisher fatalities occur despite Norway's established, regulatory framework governing fleet safety – including requirements for vessel stability, mandatory safety equipment and inspections by the Norwegian Maritime Authority. This highlighted the need to better translate these requirements into practical safety measures for everyday work practices and creating a stronger safety culture.

"Safe Fishing" is the Norwegian Society for Sea Rescue's key contribution to Norway's National Action Plan for Maritime Safety, supporting the plan's goal that no one should die or be seriously injured at sea.

Images: A small fishing vessel partially submerged in rough seas (L). Recipient of the 2025 Safe Fisher Award with their nominator (R). © The Norwegian Society for Sea Rescue

As a collaboration with authorities, fisher organizations and the fishing community – Safe Fishing uses a combination of national coordination and targeted engagement with fishing communities, including:

- social, digital and traditional media safety campaigns for fisher organizations and the wider industry;
- media reports on safety concerns and fisher stories;
- safety days and workshops for fishers in the community;
- demonstrations of safety gear and promotion of flotation devices and emergency technology;
- partnering with fisheries' schools to teach safety to future employees;
- rescue vessel crews directly engaging coastal fishers;
- annual Safe Fishing award for exemplary safety leadership.

Although in its early phase, the Safe Fishing campaign has boosted attention to occupational safety in the fishing sector. More fishers are joining safety discussions, organizations are working to develop modular safety days for fishers and nominations for the annual Safe Fishing award have risen, indicating stronger awareness of safety leadership among fishers.

Case study 6

Community co-designs and implements impactful Safe Fishing Plan, United Republic of Tanzania

This case study highlights the importance of inclusive, community-led governance for turning drowning prevention policy into practice, indicating that strengthening local governance frameworks (rather than relying on top-down regulation) can create the conditions for effective policies to be developed, implemented and enforced.

Poor community management of small-scale fisheries in the United Republic of Tanzania was compounding drowning risks through inadequate enforcement of lifejacket use; alcohol use before and during fishing; weak coordination among Beach Management Units (BMUs), community leaders and fisheries officers; and low enforcement of bylaws and guidelines.

Fishers, children and women engaged in fish processing were among those most affected. In response, the Tanzanian NGO Environmental Management and Economic Development Organization (EMEDO) worked to strengthen community governance in Musoma Rural District by improving relationships, coordination and the capacity of local actors to enforce local regulations. A participatory process began with community-level identification of drowning risk factors and other challenges at fishing landing sites. Community members, including fishers, boat owners, women fish workers, traders, religious and village government leaders, technical and government staff, police and BMUs then agreed on measures to address these risks, including penalties for non-compliance and provision for awareness-raising and training.

The process culminated in the Safe Fishing Plan, a community bylaw that has received formal approval and roll-out by the district authority.

Implementing the plan involved many key activities: capacity building through leadership training and drowning-prevention awareness sessions for fisheries officers, community leaders, BMUs and police; community awareness raising through water-safety sessions and testimonies from local drowning-prevention champions; integrating water-safety awareness into community sports activities to reach broader audiences; and enforcement and governance through BMU leaders and community leaders carrying out sensitization campaigns and safety patrols.

The plan has had a positive impact in terms of governance, behaviour change and institutional engagement at community, district and regional levels. Permanent drowning-prevention committees have been established and their reports are now fixed agenda items in both BMU and village council meetings. Community relationships have been strengthened and drowning prevention has been elevated onto the agenda of district-level management meetings.

Significantly, the Technical Management Committee of Musoma Rural District formally requested replication of the Safe Fishing Plan across all landing sites in Musoma, while the Fisheries Department in Musoma introduced a dedicated drowning-prevention section (recording reported cases and response efforts), into the standard fishing reporting template.

Images: Community members gather for a water safety awareness session, © EMEDO





T



Regulate

Transport safety standards for water travel

Mandate and enforce best-practice standards and incorporate conventions for domestic water transport safety that regulate vessel operations, availability of safety equipment and operator skills and conduct.

What to do?

- **Align** vessel safety standards with international best-practice regulations, requiring routine inspection and certification of vessel structure, maintenance and lifesaving equipment.
- **Mandate** training and certification for commercial vessel operators and crew in emergency preparedness, evacuation procedures, rescue, first aid and communications.
- **Enforce** operational systems to prevent vessel overloading, manage passenger safety, restrict operations during hazardous weather conditions and prevent alcohol- and drug-impaired operation of vessels.

Why do it?

In many settings people rely on domestic water transport to reach work and school, as well as to access critical services such as health care. Safe transport over water – including informal transport, inland waterways, ferries and small passenger vessels – underpins access to opportunities, livelihoods and social inclusion *(23, 24)*. However, all water transport poses a drowning risk if safety is not prioritized.

Vessels may be poorly maintained and uninspected, operators and crew may lack adequate training and lifesaving equipment may be insufficient, substandard, or inaccessible.

In low-income contexts, economic pressures can lead vessel operators to overload boats, operate in adverse weather, or operate unseaworthy vessels. Regulated transport that balances enforcement with practical compliance can significantly help low-income communities by reducing deaths and injuries and creating safer, more resilient transport systems *(24)*, in line with international commitments (see Box 11).

Co-benefits for communities and multiple sectors can include:

- enhanced confidence in water transport, supporting its use and benefiting local economies;
- greater equity through safe travel to school, work and essential services for all population groups;
- less disruption of daily commuting and trade in communities where water transport underpins economic and social activity;
- strengthened country reputation for safety and reliability in transport, tourism and international maritime partnerships;
- more resilient households through safeguarding transport workers' lives and livelihoods;
- reduced costs from vessel losses and insurance claims, delivering wider economic gains.

Box 11. Related international commitments

- IMO – Model Regulations on Domestic Ferry Safety: Enhance safety standards in domestic ferry operations including vessel construction, safety management, crew training and operational procedures
- IMO – SOLAS Convention: Specify minimum standards for the construction, equipment and operation of ships, compatible with their safety
- IMO – Standards of Training, Certification and Watchkeeping
- International Life-Saving Appliance (LSA) Code: Specific technical requirements for the manufacturing, testing, maintenance and record keeping of life-saving appliances
- SDG 11: Make cities (and transport systems) inclusive, safe, resilient and sustainable

How to do it

Know the risks

- Identify gaps in national and subnational laws, standards and enforcement for water transport operations, vessel design, maintenance and crew competency, including within informal and small-scale transport systems.
- Analyse incident, near-miss and inspection data to identify high-risk routes, vessels, operating conditions and seasons.
- Assess maritime, transport and port authorities' capacity to inspect, manage data and enforce regulations, including for inland waterway transport.

Make safety the standard

- Mandate regular drills and simulation exercises with maritime rescue coordination centres and port authorities to maintain readiness for mass rescues or evacuations.
- Roll out routine vessel inspection and certification for safety, maintenance and equipment compliance.
- Restrict vessel operations when adverse weather, monsoons, rough seas, or darkness exceed vessel capability.
- Implement drug and alcohol monitoring programmes for vessel operators, including random or targeted breath-testing and mandatory testing following incidents, near-misses or suspected impairment.

How to do it

Design for safety

- Incorporate SOLAS and other IMO standards (see Box 12) for passenger vessel safety into national laws, while adapting implementation approaches for inland waterways and small-scale domestic transport, where appropriate.
- Mandate routine inspection and certification for vessel structure, lifesaving gear (in line with the LSA Code, see Box 12), communication and navigation equipment and operational procedures.
- Establish certification and periodic mandatory training for operators and crew in emergency preparedness, evacuation, rescue, first aid and communications.
- Require passenger management systems to track passenger numbers as they embark/disembark to prevent overloading and enable rapid accountability during emergencies.
- Enhance access to timely weather forecasts and hazard alerts through radio, SMS or dedicated maritime communication frequencies.

Box 12. The International Life-Saving Appliance (LSA) Code

The Life-Saving Appliances (Including LSA) Code 2023 edition *(25)*, published by IMO, establishes global standards for life-saving equipment carried on vessels. It specifies the design, performance, testing and maintenance requirements for key safety systems including lifejackets, lifebuoys, life rafts, lifeboats, rescue boats, evacuation systems and emergency signals, to ensure that vessels are equipped to protect life during maritime emergencies. It is a paid-for resource available from IMO.

Keep it safe

- Mandate transparent, centralized safety audits and reporting for vessel maintenance, port operations and adverse events/near misses to guide enforcement and strengthen policy.
- Drive compliance through communication campaigns and operator engagement, emphasizing the benefits of following vessel safety standards.
- Strengthen maritime, transport and port authorities' ability to inspect, manage data and enforce compliance, supported by sufficient staff, equipment and training, including coordination across marine and inland water transport systems.

Key partners and sectors to involve

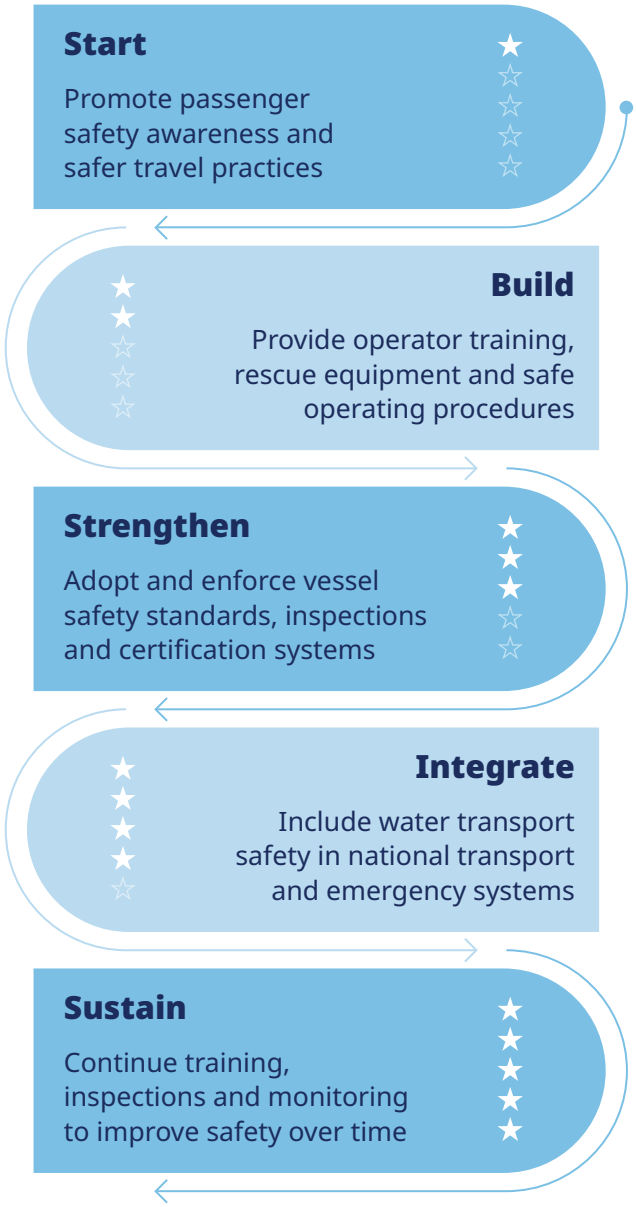
Preventing drowning through regulating transport safety standards for water travel involves many sectors and actors:

- **Government** sectors include transport, maritime safety, port authorities, coast guards, rescue coordination centres, etc.
- **Nongovernment** actors include standards bodies and certification agencies, boat-owner cooperatives and unions, private sector suppliers and vessel builders.

PROTECT pathway to progress

The pathway to progress for this recommendation – set out in Fig. 7 – is neither prescriptive nor linear but gives an example of how regulating water safety for travelling over water can evolve and save lives through awareness, training, inspections and embedding water transport safety in national emergency systems.

Fig. 7. PROTECT pathway for the "Regulate Transport safety standards for travelling over water" recommendation



Case study 7

Strengthening vessel inspection and life-saving appliance standards for domestic water transport, Indonesia

This case study demonstrates how routine vessel inspection, certification systems and life-saving appliance standards can strengthen the safety of domestic water transport and reduce drowning risk during maritime emergencies.

As the world's largest archipelagic nation, Indonesia relies heavily on domestic maritime transport to connect thousands of islands and support access to education, employment, health care and other essential services. Millions of passengers travel annually on ferries, passenger vessels and smaller watercraft operating across diverse geographic and environmental conditions. Historically, vessel overcrowding, inconsistent safety compliance, inadequate life-saving equipment and variable enforcement of maritime regulations have contributed to maritime incidents and loss of life.

To strengthen passenger safety, the Indonesian maritime administration has developed and implemented a comprehensive vessel inspection and certification system. Inspection programmes focus on vessel seaworthiness, operational safety and the availability of functioning life-saving appliances.

Passenger vessels are routinely inspected, with modern passenger vessels undergoing regular annual docking and recertification, and traditionally constructed passenger vessels inspected and recertified more frequently. Inspection systems also assess the condition, installation and type of life-saving appliances, helping ensure that equipment remains fit for purpose and available during emergencies.

This regulatory approach is complemented by public awareness campaigns and passenger safety education. On many passenger vessels, safety information and life-saving procedures are communicated to passengers before departure, including on the location and use of safety equipment. These measures are supported through national maritime safety governance systems involving vessel operators, inspectors, certification authorities and transport regulators.

This case study demonstrates that strengthening inspection, certification and compliance systems can help embed safety in routine vessel operations rather than leaving safety up to the individual passenger behaviour.



Image on next page:
Waterfront settlement with small fishing boats,
© WHO / Gary Hampton



E



Expand

Education in basic swimming and water safety

Integrate basic swimming and water safety education across the life course into school curricula and workplace and community programmes, backed by national standards and instructor certification.

What to do?

- **Institutionalize** basic swimming and water safety education within national education, workforce development and community learning systems, supported by national standards.
- **Mandate** instructor certification aligned with national standards and implement quality-assurance systems for training delivery.
- **Implement** policies and financing mechanisms that ensure equitable access to safe training environments, addressing location, gender, disability, age and financial barriers.

Why do it?

Basic swimming and water safety skills help people engage safely with water during domestic, occupational or recreational use, helping them recognize, avoid and respond safely to water hazards *(2)*. Swimming and water safety education can be taught across all age groups and adapted to different contexts and resources, making them a practical and scalable investment in public health and community resilience *(2)*. They contribute to a safer workforce in sectors that require regular exposure to water, such as fisheries, transport and tourism *(2)*.

Detailed guidance exists on how to implement safe and effective swim skills programmes (see Box 13), not only helping reduce death and injury, but also promoting child development and lifelong learning, in line with international commitments (see Box 14).

Co-benefits for communities and multiple sectors can include:

- enhanced population fitness, cardiovascular health and well-being;
- increased confidence, social connectedness and opportunities for all ages and abilities, including people with disabilities;
- stronger community capacity to support safe work practices around water and respond effectively during floods and other climate-related hazards;
- safer, more sustainable growth in water-based industries such as fishing, transport, tourism;
- reduced economic losses and medical costs due to drowning injury and death.

Box 13. Implementing basic swim-skills programmes

Basic swimming and water safety education provided to children aged six years and older can help prevent drowning by giving them simple survival skills such as floating, breath control in water, moving through water and getting themselves to safety *(26)*. Programmes should focus on these life-saving skills rather than advanced swimming, be taught safely by trained instructors and involve communities to ensure they are accepted, well run and effective *(26)*.

Box 14. Related international commitments

- Convention on the Rights of the Child: Article 6 obliges governments to ensure the child's survival and development to the maximum extent possible. Article 24 requires governments to ensure access to safe environments so that all children can grow and develop fully. Articles 28 and 29 affirm every child's right to education that supports survival skills and healthy development
- SDG 4: Ensure inclusive, equitable, quality education and promote lifelong learning opportunities for all

How to do it

Know the risks

- Assess population-level swimming and water safety knowledge (and disparities) by age, geography, gender, disability, language and socioeconomic status.
- Map current training offered through schools, workplaces, community programmes and private providers to identify gaps and integration opportunities.
- Identify participation concerns such as cost, cultural and gender norms, disability, language, facility access and safety considerations.
- Identify groups at higher risk who may have limited access to swimming and water safety education programmes, including children not attending school, or minority populations and immigrants.

Make safety the standard

- Set national, quality-assured standards for swimming and water safety education.
- Explore opportunities to utilize existing systems to deliver training: for children, this may include integration into school curricula and certification of physical education teachers as swim instructors; for workers, this may include integration into occupational health and safety programmes.
- Dedicate sustainable government funding, complemented by community co-funding and private and civil society partnerships, for scale-up and sustainability.
- Use targeted subsidies and programme adaptations to remove barriers to participation.

How to do it

Design for safety

- Use schools, community centres, private facilities (such as hotels) and natural water bodies that meet defined safety criteria to expand access to safe and inclusive training venues.
- Ensure instructors are trained, certified and supported to deliver quality-assured, safe instruction.

Keep it safe

- Establish monitoring and quality-assurance systems to track participation, safety and outcomes, using data to improve training design.
- Where possible, use existing monitoring systems in the education, labour or health sectors to track programme quality and avoid duplication of reporting processes.
- Strengthen alignment with national education and health-reporting frameworks.
- Periodically review standards, curricula and training delivery to ensure continued relevance and effectiveness.

Key partners and sectors to involve

Preventing drowning through expanding education in basic swimming and water safety involves many sectors and actors:

- **Government** sectors include health, education, sport/youth/recreation, labour and local government agencies.
- **Nongovernment** actors include sports federations, swimming associations, pool operators and leisure facilities and volunteer groups.

PROTECT pathway to progress

The pathway to progress for this recommendation as set out in Fig. 8 (which is neither prescriptive nor linear) aims to show how expanding education in basic swimming and water safety can develop over time to strengthen knowledge and skills in all segments of the population and embed the concept at all levels that safety is paramount around water.

Fig. 8. PROTECT pathway for the recommendation "Expand Education in basic swimming and water safety"



Image on next page: Children participate in structured swimming and water safety classes.
© WHO / Ho Chi Cuong



Case study 8

Survival swimming for children aged 6–15 years, Viet Nam

This case study illustrates how Viet Nam has institutionalized child drowning prevention through the adoption of major national policies, marking a transition from pilot interventions to nationally endorsed, system-embedded prevention measures, long-term sustainability and scale-up.

Drowning remains the leading cause of child mortality in Viet Nam, with limited access to water safety education and survival swimming skills among children a key contributing factor (27). Each year around 1800 children die from drowning. To address this, in 2018 government ministries, international organizations and WHO developed a national guide and curriculum for survival swimming and water safety education. Pilot interventions were implemented in eight high-burden provinces, providing structured survival swimming and water safety classes for children aged 6–15 years.

Investment in programme delivery was prioritized. This ensured that managers, government officials and swimming instructors were equipped with the technical knowledge to ensure safe and effective instruction, while collaboration mechanisms and capacity-building activities further reinforced shared accountability among the health, education, social affairs, sports sectors and community engagement. A robust monitoring and evaluation system, quality assurance mechanisms and structured case documentation were also established to capture essential information on programme delivery, quality and effectiveness.

By 2025, more than 400 000 children in Viet Nam had completed survival swimming training. At intervention sites, the proportion of children aged 6–15 years able to swim increased from 15% to 46% over eight years, exceeding the national average.

In addition, 52 250 children gained essential water safety knowledge, 1588 instructors were trained to international standards and 2250 officials from the health, education and community sectors completed structured drowning prevention capacity-building programmes. These achievements helped strengthen cross-sectoral coordination and support the sustainability of the intervention.

The programme – later expanded – has ultimately led to a 16% reduction in drowning rates among children aged 6–15 years and an estimated 24 child deaths averted each year. It showed a 20-fold return on investment, making it one of Viet Nam's highest performing social welfare interventions. Survival swimming and water safety education are now integrated into the country's national child injury prevention framework, education system and child rights agenda.

Viet Nam's experience shows that child drowning prevention can move beyond short-term projects when supported by long-term policy, local ownership, standardized techniques, cross-sectoral coordination and sustained funding.

Images: Children participate in supervised swimming and water safety instruction, © GHAI Viet Nam







Case study 9

Swim and Survive – a nationwide water safety programme, Australia





This case study demonstrates how embedding swimming and water education within policy systems (health, sport, industry) can support population-level drowning prevention.

Despite Australia's strong aquatic safety culture, drowning remains a leading cause of unintentional child death in the country. Swim and Survive, established in 1982 by Royal Life Saving Society – Australia, initially targeted children aged 5–14 years, later expanding to meet the needs for children under the age of 5 years, as well as teenagers (due to increased drowning risk among those aged 10–20 years), late learners and adults.

Delivered nationally through partnerships with schools, aquatic facilities, local government and community organizations, the programme:

- sets nationally consistent swimming and water safety benchmarks and standards;
- integrates water safety into school-based education systems;
- reduces inequities in access across states and territories;
- improves water safety outcomes; and
- aligns with national drowning prevention strategies and actions.

Swim and Survive aligns with Australia's National Swimming and Water Safety Framework, providing a national, standardized swimming and water safety curriculum spanning early childhood to adolescence and integrated into school and community aquatic programmes. It has also established competency benchmarks for skills such as distance swimming, CPR, float survival, safe rescue; provided instructor training and standards; adopted inclusion strategies for vulnerable populations; and developed national quality assurance and monitoring systems.

Data show that, to date, the programme has contributed to sustained reductions in child drowning rates, demonstrating that embedding swimming and water education within policy systems can prevent drowning and that standardized curricula can improve consistency, quality and equity. It also shows that an NGO can act as a national focal point for coordinating multisectoral implementation and that responsibility can be shared across the education, health, sport and industry sectors. Finally, the Swim and Survive story shows that sustained investment in structured swimming and water safety education is required for long-term reductions in child drowning deaths.

Images: Children learn swimming and water safety skills under the guidance of trained instructors, © RLSS Australia

C



Reinforce

Child-care systems and ensure safeguarding

Ensure preschool children have access to appropriate care with effective adult supervision across all care systems, protecting them from drowning risks and promoting safe, healthy development.

What to do?

- **Institutionalize** government-supported supervised child-care services as a core drowning prevention measure within national child welfare and early childhood development systems.
- **Mandate** safeguarding and environmental safety standards that reduce young children's exposure to water hazards in homes, child-care settings and public spaces.
- **Require** drowning prevention to be incorporated into caregiver education, parenting programmes and early childhood development frameworks.

Why do it?

Proper adult supervision lowers child drowning risk and enhances children's safety, health and development *(28)*. Ensuring such supervision in all child-care settings (see Box 15) is vital for reducing exposure to water-related risks and raising water safety awareness among children, families and communities. Drowning prevention can be integrated into early childhood development frameworks.

Quality child care not only helps prevent child deaths but also promotes early childhood development and protects children's rights under the UN Convention on the Rights of the Child (see Box 16).

Co-benefits for communities and multiple sectors can include:

- lifelong water safety awareness and behaviours;
- strengthened child protection, early childhood development and equitable access to supervision and care within the community;
- improved school readiness, higher educational achievement and reduced school drop-out;
- enhanced family well-being and safer home and community environments, reducing risk of neglect and unintentional injury; strengthened intergenerational awareness and engagement among children, parents and caregivers;
- stronger family and community resilience, particularly during seasonal events or flooding;
- reduced burden on health, emergency and social services, improving community well-being;
- greater employment or education opportunities for caregivers (often women) to support household income, promoting gender equality.

Box 15. Defining child care

Child care varies widely across contexts, shaped by society's norms, community structures and the availability of resources and facilities. Care for young children can be through formal or informal structures – from family care provided by parents, grandparents or neighbours, to centre-based care provided by trained child-care workers; or home-based care in the child's home (e.g. nannies) or in a practitioner's home (e.g. a childminder).

Box 16. Related international commitments

- Convention on the Rights of the Child: Articles 18 and 28 oblige governments to provide appropriate assistance to parents and ensure children's right to education
- SDG 4.2: Ensure that all girls and boys have access to quality early childhood development, care and pre-primary education
- UN Committee on the Rights of the Child: Emphasizes governments' duty to make early childhood care services universally available and accessible

How to do it

Know the risks

- Identify locations where children regularly access or spend time near water, including play areas located near open water or flood-zones.
- Assess child drowning risk across all care settings, identifying when and where children may be exposed to water.
- Map current child care and supervision practices, identifying gaps in access to formal child care and reliance on informal or community-based care.
- Review regulations, workforce capacity and data from monitoring systems to identify strengths and gaps.

Make safety the standard

- Apply minimum, context-appropriate safety and supervision standards in all child-care settings to strengthen them over time as resources, staff skills and systems improve.
- Embed water safety criteria into child-care licensing and accreditation, including requirements for protective infrastructure, safe facility design and clear supervision protocols.
- Mandate qualification and training requirements for child-care staff, including knowledge of drowning prevention, water-risk management, rescue readiness, first aid and appropriate adult-to-child supervision ratios.

How to do it

Design for safety

- Train parents, caregivers and child-care providers in appropriate supervision practices, drowning awareness, first aid, safe rescue and CPR, using scalable, appropriate training developed together with communities.
- Integrate drowning prevention messaging into community health, early childhood development and social protection platforms, ensuring consistent and reinforced messaging.
- Establish supervised play areas in communities near water, with shade, seating, clear signage and lifesaving equipment.
- Make child-care centres safer by installing barriers around nearby water, creating controlled entry and exit points and improving drainage where needed.

Keep it safe

- Mandate safety audits of day care and early childhood facilities, especially near water, to identify and address hazards.
- Implement supportive (not punitive) regulatory and monitoring systems that enable incremental safety improvements, particularly in low-resource settings.
- Maintain training and refresher programmes for child-care staff and community caregivers.
- Reinforce drowning prevention messaging during routine contact with families and caregivers via health, education and social services.

Key partners and sectors to involve

Preventing drowning through strengthening child supervision involves many sectors and actors:

- **Government** sectors include education, health, social welfare/family services and rural development.
- **Nongovernment** actors include private sector child-care providers, parents' associations, community leaders.

PROTECT pathway to progress

The pathway to progress for this recommendation as set out in Fig. 9 (which is neither prescriptive nor linear) aims to show how reinforcing child-care systems and ensuring safeguarding can evolve, reducing drowning risk for those most vulnerable to it.

Fig. 9. PROTECT pathway for recommendation "Reinforce Child-care systems and ensure safeguarding"

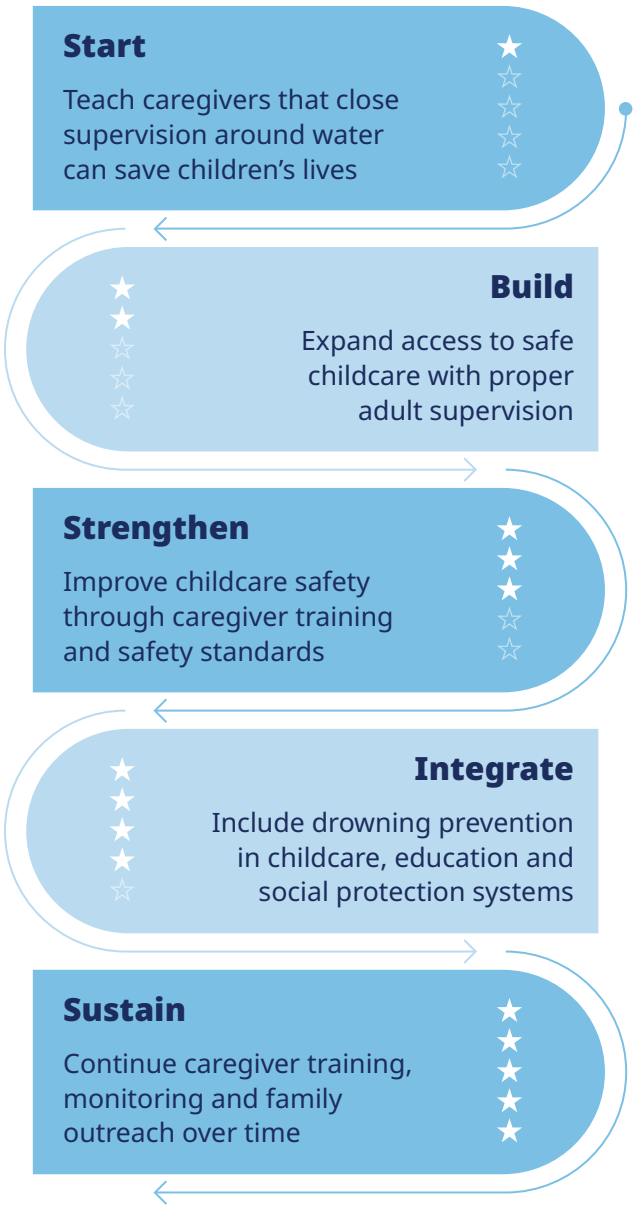


Image on next page: Early childhood educator supervising young children during outdoor play. © WHO / Alex Swanepoel



Case study 10

Preventing drowning through state-supported child-care systems, Thailand

This case study demonstrates how government policies that set standards for child development centres and integrate water safety can protect children's rights, reduce safety inequities (including drowning risk) and establish system-level protection for children.

In Thailand, water safety has been formally embedded in the early childhood care system since 2016, through the integration of drowning prevention with the operational standards of child development centres. These standards require safe environmental management in and around the centres, including eliminating open-water hazards and providing safe sanitation.

This policy measure, mandated by national government for nationwide implementation and delivered through government-supported child development centres, works as a social protection mechanism, reducing the caregiving burden on (especially older) family members and supporting safe and healthy child growth and development. It does this by requiring all centres to ensure that environments, infrastructure and child-care systems comply with the standards and tasking local administrative organizations to oversee, supervise and enforce them.

The measure is supported through the government programme known as MERIT MAKER, an acronym that stands for 10 strategies (management, environment, resuscitation, information, training (survival swimming), media, advocacy, kindergarten/day care, education and research).

The programme builds teacher and caregiver capacity to improve child-care environments to address water-related risks and provide ongoing supervision.

As a systematic and sustainable mechanism to link national policy with local-level practice, the programme offers practical implementation tools and encourages monitoring to ensure consistent implementation across child-care centres.

Monitoring data from centre supervisory systems and from MERIT MAKER itself show no reported drowning incidents in centres implementing the prescribed standards and that drowning deaths among children aged 2–5 years nationally have fallen 24%. Young children also demonstrate increased knowledge and awareness of drowning risks, can identify unsafe behaviours and environments and actively transfer this knowledge to their families – encouraging parents and caregivers to improve water safety conditions at home.

This case study demonstrates that national policies and standards can shift drowning prevention from being an individual responsibility to being a system-wide safety approach by embedding child safeguarding and water safety measures within the child-development centre system. This results in risk reduction that is comprehensive, sustainable and extendable to families and communities.

Image: Young children learn about water safety in classroom, © Thai Ministry of Public Health



Case study 11

Strengthening local government capacity to support child supervision and drowning prevention, Philippines



This case study demonstrates how a nationally led capacity-development programme can strengthen local government systems to help parents and caregivers protect young children from drowning.

In the Philippines, drowning prevention has been integrated into local health-promotion systems through the Promoting actions toward drowning-free communities playbook *(29)*, which provides local government units (LGUs) with practical guidance, tools and resources to plan, implement, monitor and sustain community-based drowning prevention initiatives. Recognizing that children aged 0–4 years are at greatest risk of drowning, the playbook (developed by the Philippines' Department of Health, in partnership with WHO and national partners) emphasizes supporting caregivers, strengthening supervision and increasing community awareness of childhood drowning risks.

The playbook uses a cascading capacity development approach to equip local leaders, neighbourhood health workers and community volunteers with the knowledge and skills needed to implement drowning prevention measures.

Through standardized training materials, communication resources, policy templates and monitoring tools, it supports consistent implementation across diverse local contexts while allowing LGUs to adapt activities to local needs and risks. Suggested activities include caregiver education sessions, mothers' classes, community assemblies and public awareness campaigns that reinforce effective supervision and safe environments for young children.

The programme promotes a multisectoral approach by engaging health, education, social welfare, disaster risk management and community development sectors in a shared effort to protect children. It is supported by a monitoring and evaluation framework linked to the national drowning prevention strategy, enabling LGUs to track progress using practical indicators integrated into existing systems.

Image: *Young children play near river in rural community, © WHO / Yoshi Shimizu*

T



Improve

Threat preparedness for floods and other multi-hazard emergencies

Legislate for comprehensive multi-hazard preparedness systems that enable coordinated early action, response and recovery across sectors.

What to do?

- **Establish** coordinated governance and institutional mandates for multi-hazard preparedness that define clear roles, responsibilities and standard operating procedures for early action, response and recovery.
- **Mandate** the development, operation and maintenance of national multi-hazard early-warning systems that provide timely, accessible and actionable risk information.
- **Establish** financing mechanisms and operational frameworks that support community-level preparedness and emergency response capacity.

Why do it?

Floods are among the most widespread and damaging natural hazards, affecting communities in all regions of the world *(30)*. Floodwaters can move rapidly, conceal hidden debris and remain in communities for weeks, prolonging the risk of drowning, which accounts for approximately 75% of deaths during flood-related disasters *(31)*.

In many settings, climate change is altering the frequency, intensity, timing and geographic distribution of floods and other weather-related hazards *(30)* creating new challenges for preparedness and response. In addition to floods, drowning and water-related injury risks can arise during storms, high winds, ice and snow events, often with cascading impacts on health, infrastructure, transport, livelihoods and essential services.

Effective multi-hazard preparedness – enabling rapid, safe and coordinated community response and supporting recovery – requires coordinated action across sectors and between countries, with clear roles and responsibilities at national and local levels. Investing in preparedness reduces death (including from drowning), safeguards livelihoods, public health and socioeconomic resilience during emergencies and enables communities to build back better after disasters. It also helps countries meet multiple international commitments (see Box 17).

Co-benefits for communities and multiple sectors can include:

- strengthened community capacity to act collectively and confidently before, during and after emergencies;
- improved household resilience, enabling families to protect assets, safeguard livelihoods and recover more quickly, reducing economic disruption;
- decreased pressure on emergency services, health systems and humanitarian response mechanisms;
- sustained social well-being and economic productivity from minimized injury, displacement and work interruption;
- stronger public trust in government, encouraging people to follow preparedness measures and official safety guidance.

Box 17. Related international commitments

- Sendai Framework for Disaster Risk Reduction (Priorities 1–4): Understanding, strengthening governance around, investing for resilience in, and enhancing preparedness for disaster risk
- Paris Agreement (Article 7): Understanding climate change adaptation and defining a framework for action
- SDGs 1.5, 3.d, 11.5 and 13.1: Build resilience to environmental, economic and social disasters; strengthen countries' capacity, in particular developing countries, for early warning, risk reduction and management of national and global health risks; Make cities (and transport systems) inclusive, safe, resilient and sustainable; Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters, respectively
- United Nations Framework Convention on Climate Change: Acknowledge the existence of the threat of climate change and agree to undertake efforts to combat it

How to do it

Know the risks

- Acknowledge that drowning is a predictable and preventable risk during floods and other extreme weather and climate-related hazards and review disaster plans to ensure they address drowning prevention, rescue and response.
- Analyse multi-hazard risk, exposure and vulnerability using data from meteorological offices, water authorities and disaster-management agencies.
- Review institutional mandates and coordination arrangements across regional, national and local emergency response agencies to identify gaps, overlaps or bottlenecks.
- Map community preparedness, including access to information, local response mechanisms and existing partnerships with authorities across all local, regional and state borders.
- Review existing disaster management systems, community-based disaster risk management structures and local preparedness committees to identify opportunities to strengthen coordination, improve interoperability, accelerate implementation and support sustainable, locally owned preparedness efforts.

Make safety the standard

- Establish clear institutional mandates for multi-hazard preparedness, response and recovery across all levels of government.
- Mandate joint planning and information-sharing between meteorological services, disaster management agencies, telecommunication providers, water authorities and local, regional and state governments, ensuring timely information is shared with at-risk groups. This includes sharing hazard-risk modelling and data to inform cross-sector decisions.
- Legislate for weather and hazard warnings that use accessible language, channels and formats to communicate timing, severity and advice; and that reach populations with poor literacy and access to technology through community mechanisms, with system-failure backup (see Box 18).
- Develop anticipatory funding for measures such as barriers, evacuation drills and temporary asset protection, with localized action grounded in government guidance.
- Align national preparedness, response and recovery standards with best-practice frameworks (e.g. the Sendai Framework) to support cross-border cooperation during major floods and other climate-related emergencies and enable coordinated assistance when national response capacity is exceeded.

Box 18. End-to-end early warning systems

Effective end-to-end early warning systems should translate technical forecasts into clear, timely and actionable information for communities. Early-warning messages should clearly describe the type and location of the hazard, when it is expected to occur, how severe it may be and which people, areas or infrastructure are at risk. They should also provide clear protective actions, identify the official authority issuing the warning and indicate where further information or updates can be accessed through trusted communication channels.

Effective systems should include impact-based forecasting, risk communication, last-mile dissemination and feedback mechanisms to ensure warnings are understood and acted upon. Community-based early warning systems – including through community-based disaster risk management platforms and committees, local volunteer networks and local government preparedness structures – can help strengthen preparedness, local ownership and rapid community response *(32)*.

Design for safety

- Empower communities through participatory risk assessments, awareness campaigns, training exercises and preparedness meetings that reflect differing levels of risk and capacity, ensuring tailored early protective actions for groups such as older people, children and caregivers and people with disabilities. Identify and address the reasons why communities or individuals may not move or take the necessary action in response to a hazard warning.
- Coordinate local community task forces to support joint planning, mobilization and response, guided by national and regional authorities.
- Ensure disaster-response plans include hazard-specific drowning prevention, evacuation and rescue measures, scaled to event severity.
- Promote nature-based solutions as part of multi-hazard preparedness and risk reduction strategies, including watershed management, mangrove restoration, urban green infrastructure, slope stabilization and ecosystem-based flood buffering to reduce hazard exposure and strengthen climate resilience.

How to do it

Keep it safe

- Post-disaster, prioritize resilient reconstruction, nature-based solutions and community-led adaptations that reduce future exposure and vulnerability.
- Review response performance and outcomes to identify lessons learned, including coordination effectiveness, warning reach and rescue capacity.
- Use recovery processes to strengthen long-term preparedness systems, embedding improvements in plans, financing mechanisms and institutional mandates.
- Maintain cross-border and regional cooperation mechanisms to sustain collective readiness.

Key partners and sectors to involve

Drowning prevention through multi-hazard preparedness involves many sectors and actors:

- **Government** sectors include meteorological services, disaster risk management, telecommunications regulators, utility providers, local government units, fire and civil defence and the military.
- **Nongovernment** actors include telecommunications and Internet service providers, national and local media (TV, radio, newspapers, community broadcasters), community leaders, search and rescue and lifesaving associations.

PROTECT pathway to progress

The pathway for the recommendation "Improve threat preparedness for floods and other multi-hazard emergencies", as set out in Fig. 10 (which is neither prescriptive nor linear) aims to show how preparedness can be built, strengthened and integrated over time, making populations safer during times of emergency.

Fig. 10. PROTECT pathway for recommendation "Improve Threat preparedness for floods and other multi-hazard emergencies"



Case study 12

Strengthening disaster-risk knowledge through citizen science, Brazil

This case study demonstrates how involving communities in producing flood-risk information can strengthen disaster-risk knowledge, improve preparedness and build trust in national early warning systems.

In Brazil, recurrent flooding affects many urban communities, particularly those in informal settlements and other flood-prone areas. Traditional disaster-risk information systems often rely on technical data collected by specialist agencies, with communities primarily receiving warnings after risks have been identified. Recognizing the value of local knowledge and lived experience, the Dados à Prova d'Água (Waterproofing Data Project) was developed to enable citizens to actively contribute to generating flood-risk information.

The initiative engaged teachers, students, civil defence personnel and community members across more than 20 municipalities in five Brazilian states. Through a citizen-science approach, participants worked together to collect, validate and communicate local flood-risk information using tools designed collaboratively with them. The project particularly engaged school-aged children and young people, encouraging intergenerational participation and capturing local knowledge and community memory from a diverse range of perspectives.

A key feature of the programme was its integration with Brazil's National Center for Monitoring and Warning of Natural Disasters (CEMADEN). Citizen-generated information was combined with technical monitoring systems to support hazard assessment and generate warnings, creating stronger connections between communities, scientists and government agencies. Educational materials and learning guides helped participants understand flood processes, risk factors, data collection methods and disaster risk management, strengthening community awareness and preparedness.

By involving communities directly, the initiative helped shift citizens from being passive recipients of warnings to active contributors to disaster risk management. The programme also strengthened trust in official warning systems, fostered local ownership of preparedness activities and created new partnerships between communities, government agencies and technical experts. Importantly, it demonstrated that locally generated knowledge can complement scientific data and improve the relevance and usability of risk information. This case study shows how empowering communities to generate, interpret and use hazard information can strengthen disaster-risk knowledge, support more inclusive early warning systems and improve resilience to flooding and other climate-related hazards.

Image: Students in Santa Catarina, Brazil using the Waterproofing Data mobile phone application, © Waterproofing Data Project



Case study 13

Empowering professional and community response in emergencies, Pakistan

This case study describes how a nongovernmental disaster risk management programme in Pakistan (PEER), has evolved over time to institutionalize threat preparedness at all levels, including government, equipping the population with the knowledge and skills to save lives through coordinated disaster response.

Between 2010 and 2020, Pakistan experienced 211 disasters that affected more than 51 million people and claimed over 11 000 lives. From 2008 to 2021, floods caused over 90% of disaster-related displacement. These impacts highlight the need for resilient infrastructure and a comprehensive approach to disaster risk management.

Pakistan's disaster preparedness approach reflects its regional diversity, with strategies adapted to local hazards, capacities and operational realities. This has encouraged stronger local ownership and improved coordination between national and district institutions.

A central pillar of Pakistan's disaster risk management is the Programme for Enhancement of Emergency Response (PEER). Launched by the United States Agency for International Development (USAID) in 1998 and adopted by Pakistan in 2007 after the 2005 Kashmir earthquake, PEER is a multi-hazard capacity-building initiative that strengthens preparedness and supports integrated emergency management at national, subnational and community levels.

By fostering local ownership, PEER has developed practical strategies suited to Pakistan's diverse risks, institutions and community needs. It has also sustained emergency response training and enabled organizations to train personnel to internationally recognized safety and response standards.

PEER has strengthened both professional response capacity and community resilience through training and local engagement. Between 2007 and 2021, more than 25 000 people were trained through PEER, alongside an estimated 150 000 community workers. Training covers search and rescue, medical first response, hospital preparedness and community-based response. Through initiatives such as Community Action in Disaster Response (CADRE), locally trained responders act as preparedness advocates and frontline leaders. These efforts have produced practical results, including the rescue of 26 drowning victims in the river Ravi in north-east Pakistan.

In some cases, emergency response times have fallen to as low as seven minutes and reported outcomes include lives saved through CPR, rapid medical intervention and community-led rescues. Other achievements include the introduction of response standards, a national baseline study and stronger links with academic institutions. Women's participation in disaster response has also increased and community volunteers continue to play an essential operational role, including in river rescue and evacuation efforts.

PEER's experience points to a strong foundation for long-term resilience, with training now embedded in key emergency response institutions, including Rescue 1122, the Pakistan Red Crescent and the armed forces. Course accreditation and alignment with national policy frameworks have strengthened sustainability. Institutionalization now includes integration into disaster response plans and formal training curricula, while regional knowledge-sharing and South-South cooperation remain important for building resilience across the region.



Image on next page: Emergency responders practice resuscitation techniques during a simulated rescue exercise, © Rescue 1122


RESCUE
1122





Section

2

PROTECT implementation

A group of small children play on sandbags near a river, © WHO / Yoshi Shimizu

PROTECT implementation

Drowning is a complex issue requiring multiple sectors to effectively address the broad range of drowning risk factors. All stakeholders and sectors will have their own jurisdiction, knowledge and resources to contribute, and where aims intersect, each can help others achieve their goals. This "whole-of-government" approach is fundamental to all phases and steps in the drowning prevention journey and in implementing all seven PROTECT policy recommendations.

Putting the package to work

This section sets out a systems-oriented approach for implementing the PROTECT technical package. The key steps in this approach (Fig. 11) are most impactful when they involve strong public-private partnerships between multiple government sectors, civil society, academia and other nongovernmental actors.

Step 1: Review existing policies

Policy-makers can review their country's existing drowning prevention policies and laws and compare existing national drowning prevention activities to the recommendations in Section 1 of this package. This can help identify gaps and opportunities.

Step 2: Position drowning prevention as beneficial for multiple sectors

Multisectoral collaboration for drowning prevention requires policy-makers across government sectors (and in nongovernment sectors) to understand the issue and recognize that addressing it can jointly benefit the goals and mandates of all stakeholders.

Policy-makers can start with identifying and understanding the entry points for drowning prevention in the mandates of multiple government departments and sectors and setting out the contribution drowning prevention can make to achieving adjacent or cross-cutting priority agendas (thereby addressing any concerns about competing priorities).

For example, policy-makers can highlight the potential of drowning prevention as a nation- and community-building activity that contributes to overall health and well-being, social cohesion, resilience, productivity and development. They can also cultivate a shared narrative across sectors to reinforce collective ownership and accountability for drowning prevention.

Different drowning risks require different policy solutions. Together, they form a comprehensive approach to drowning prevention.

Step 3: Gather data on drowning burden

Gathering timely and accurate data on the existing drowning burden and risks is essential to gain a clear understanding of the scale, context and determinants of drowning that can make a compelling case for investing in policy. Recommendations should be closely aligned with robust, locally relevant evidence to give decision-makers confidence that proposed actions will deliver measurable impact.

For newly proposed policy interventions, recommendations should be grounded in robust evidence of effectiveness. Where data are limited, pilot studies can be run to help demonstrate feasibility, generate local evidence and build momentum for national scale-up. Technical alignment and strong leadership can also help build the case, fostering shared purpose across sectors, helping to navigate power dynamics and encouraging adaptive collaboration to reduce resistance, secure policy approval and sustain multisectoral action.

Step 4: Build the case for drowning prevention

Building a clear case for how drowning prevention can support wider national and global priorities can increase engagement across government. This might involve demonstrating how drowning prevention strategies can help create safe and enabling environments for young children and their development, make communities more climate-resilient and reinforce progress towards broader sectoral goals and targets (e.g. reducing child mortality or strengthening the economy of water-based industries).

Policy-makers can highlight the cost-effectiveness of drowning prevention. For example, proven interventions such as supervised child care for pre-school children and swimming and water safety education for school-aged children deliver strong returns on investment, with modelled estimates suggesting that scaling these measures globally could yield savings exceeding US$ 400 billion and a nine-fold return by 2050 *(7)*.

Step 5: Select recommended policies

Policy-makers can now select from the recommendations in Section 1 of the PROTECT technical package that are most appropriate to their setting and sector/s. If their country already has a national drowning prevention strategy, it can help strengthen policy coordination across ministries and agencies. In contexts where coordinated implementation is not feasible, this package can still be used to help individual sectors take recommended actions that contribute to broader national progress on drowning prevention.

Step 6: Design policy that works across the system

Clear, feasible and context-appropriate policies that ensure drowning prevention is relevant, inclusive and effective across diverse contexts are most likely to secure funding in ways that ensure policy continuity and long-term implementation.Policy-makers should apply "systems thinking" when designing new drowning prevention policy, considering all actors and sectors that influence drowning risk (and not just a single government department – see Section 3 on adopting a systems-oriented approach to drowning prevention).

A system-wide, national drowning prevention strategy must be multisectoral, integrating complementary policy interventions across sectors and setting out clear roles (including focal points, timelines, resources and coordination/ accountability mechanisms) involving all key actors. National policies should have the flexibility to adapt to subnational contexts, capacities and local risk profiles. Where appropriate, policy-makers should strengthen or adapt existing policies rather than create new ones, maximizing efficiency and accelerating implementation. Remember – placing people at the centre of the policymaking process (through empathy-driven research, co-creation with affected communities and iterative testing) can strengthen policy inclusivity, reach and the extent to which affected communities' day-to-day risk factors for drowning are met, as well as gaining community buy-in for drowning prevention measures.

Step 7: Coordinate for effective policy roll-out

A planned and coordinated approach to implementation – outlining timelines, budgets, milestones and risk management plans – can support effective policy roll-out by all stakeholders. Clear and consistently communicated policy objectives can ensure stakeholders have a shared understanding of what is required across ministries, sectors and subnational authorities during and after policy roll-out.

Policy-makers can leverage strategic opportunities such as national planning cycles, budget reviews, or observance days to maintain political momentum and reinforce cross-sectoral commitment. Support from the public and media can be garnered through transparent and proactive communication to strengthen trust and visibility of government action on drowning prevention.

Step 8: Review, learn, and sustain

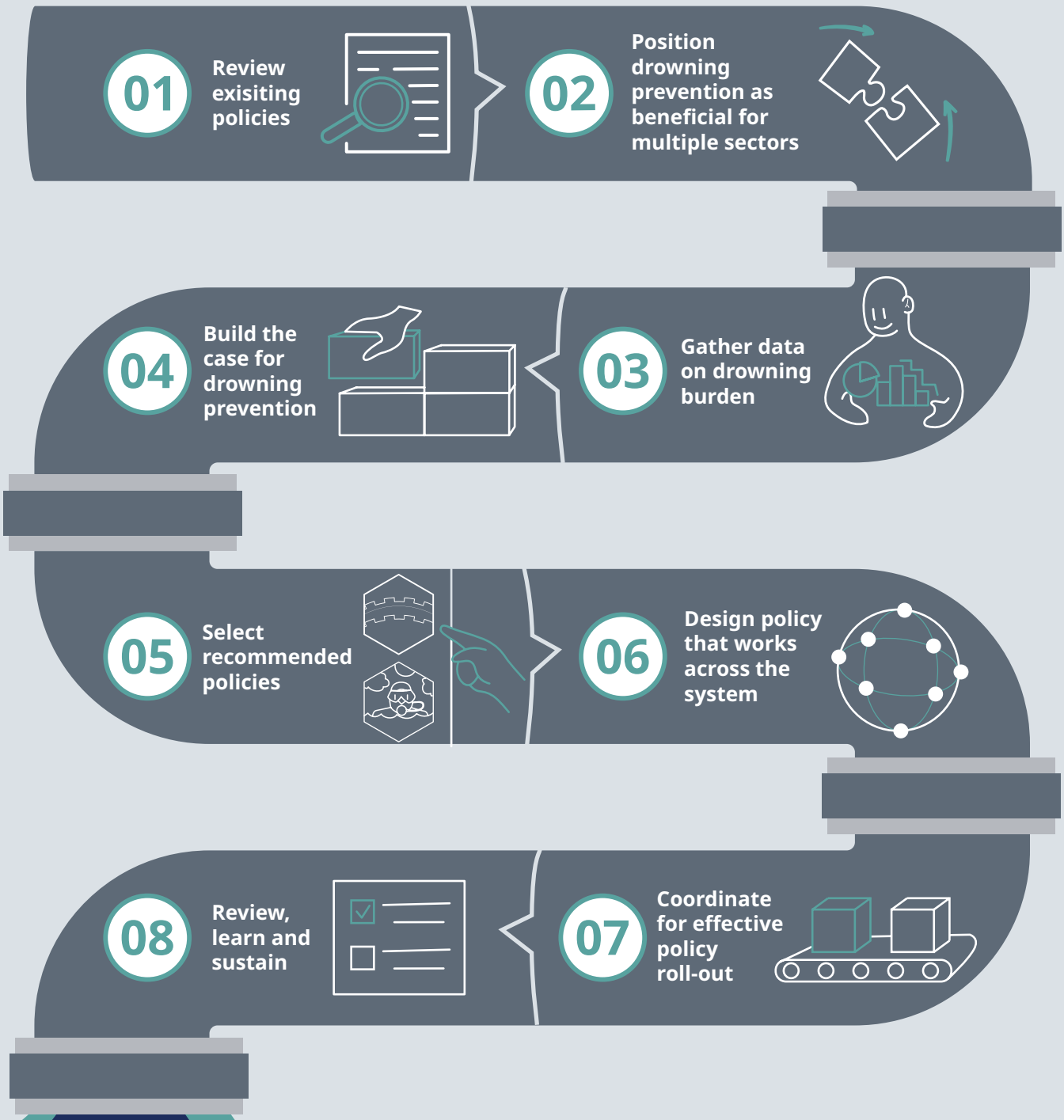
Policy-makers can monitor and evaluate policy interventions using core quantitative indicators to assess policy impact and identify and address emerging gaps and implementation challenges. These indicators can be developed using the many actions and activities set out in the PROTECT recommendations as a guide.

Robust monitoring and evaluation systems help track policy implementation, measure progress and inform adaptive decision-making. These systems should draw on existing routine data sources where possible, monitoring a concise set of high-value, measurable indicators that provide the most meaningful, reliable and actionable information on whether a policy is being implemented effectively and achieving its intended outcomes.

Policy-makers can ensure that mechanisms are in place for periodic policy review and revision based on routinely collected data, lessons learned and evolving national priorities. Whole-of-society engagement can be encouraged by enabling community participation in policy implementation and feedback processes, strengthening ownership and sustainability.

Continuous learning, collaboration and adaptive capacity can be built into implementation systems, enabling policies and programmes to evolve in response to local context, feedback, shocks and emerging risks. Sustained impact depends on the ability of institutions and communities to solve problems and adapt and strengthen relationships over time – not solely on static policy design or funding levels.

Fig. 11. Activities to position, build, design, implement and monitor drowning prevention policy



Cross-cutting components



National strategy for drowning prevention



National coordination mechanisms for drowning prevention



Appointment of national drowning prevention focal point



Strong surveillance data

Cross-cutting enablers for effective collaboration

Implementing the recommended PROTECT actions in this technical package requires supportive national systems and structures to ensure effective, systematic and robust roll-out, driven by multisectoral buy-in. This section describes four enablers that provide the systemic foundation for coordinated, sustained implementation.

1. Appoint a national focal point

A national focal point - whether a government agency, national coordinating body or designated position - provides leadership and supports coordination and accountability for drowning prevention. They can act as the driver for this technical package, support uptake of its recommendations, coordinate across sectors and ensure that actions are translated into measurable results. This role also works to ensure that progress is monitored and that the visibility of drowning prevention is maintained in broader health and development agendas.

2. Establish a national coordination mechanism

A national, multisectoral coordination mechanism formalizes collaboration among government ministries, agencies and nongovernmental partners, aligning policies, resources and activities to reduce duplication and promote shared responsibility and ownership of drowning prevention efforts. It offers a platform for joint planning, resourcing and monitoring, ensuring implementation is coherent and sustained across sectors.

3. Develop a national drowning prevention strategy

A national drowning prevention strategy provides a framework to integrate and operationalize policy actions, prioritize interventions, assign roles and responsibilities across partners and set measurable targets. As a potential roadmap for putting this package into practice, a national strategy can align interventions with national priorities, establish accountability mechanisms and provide the basis for monitoring progress towards national and global drowning prevention goals.

4. Collect population-wide drowning surveillance data

Population-wide drowning surveillance provides the necessary evidence to inform policy-makers about who is at risk of drowning and under what circumstances and to enable trends to be monitored over time. Existing health, civil registration, disaster, transport or law enforcement data systems can be strengthened to capture key drowning indicators, supporting the ongoing refinement, implementation and enforcement of policies and interventions for maximum impact.

Image on next page: *Lifeguards set warning signboards at beach, © WHO / Kimimasa Miyama*



Effective implementation depends on supportive national systems, strong multisectoral collaboration and sustained political commitment.


WARNING
NO
SWIMMING
遊泳禁止





Section 3

Systems-oriented approaches to drowning prevention

Young children fishing in river using traditional woven fish trap, © WHO / F Lang

Systems-oriented approaches to drowning prevention

The PROTECT technical package supports a systems-oriented approach to drowning prevention, recognizing that effective prevention depends on coordinated action across multiple sectors rather than delivered through isolated activities. Across the seven PROTECT recommended actions, opportunities exist at every stage of system development to strengthen safety around water, reduce drowning risk and improve long-term community resilience to the threat that water (and other hazards that intersect with it) can bring.

How a systems-oriented approach can evolve

As countries strengthen drowning prevention efforts over time, systems often evolve from isolated activities toward more coordinated, integrated and sustainable approaches (Fig. 12). This pathway reflects several core principles of systems-based approaches to injury prevention, including shared responsibility for safety; proactive risk reduction; multisectoral coordination and building environments; and systems that reduce the consequences of human error.

Applying these principles to drowning prevention means recognizing drowning as a systems-level issue that cuts across sectors, from health, education, transport and child care, to labour, planning, disaster preparedness and climate resilience. Preventing drowning therefore requires complementary and mutually reinforcing actions across policy, infrastructure, services, regulation, education and community systems.

Over time, stronger coordination, governance and institutionalization can help ensure that drowning prevention efforts become more sustainable, equitable and effective. National strategies, multisectoral coordination mechanisms and designated focal points can support alignment across sectors and help embed drowning prevention in all relevant sectoral agendas.

Evolving a systems-oriented approach to drowning prevention is unlikely to be linear – countries may be at different stages for different PROTECT domains depending on their context, capacities, priorities and existing systems. The PROTECT pathways included throughout this technical package are therefore intended as practical and illustrative examples of how systems that prioritize safety may evolve, rather than as prescriptive stages or a formal scoring framework.

The greatest impact is achieved when complementary interventions are implemented together across multiple sectors.

Co-benefits of a systems approach: a strong investment case

As the seven PROTECT technical package recommendations are inherently interconnected, progress in one area can drive progress in others, generating multiple co-benefits that reinforce the overall drowning prevention system, simultaneously advancing multiple national priorities (Fig. 12). When viewed through this lens, investment in drowning prevention becomes an investment in public health, sustainable development and social and economic resilience.

This cross-cutting approach increases the sustainability and scalability of recommended policy interventions and encourages stronger ownership across government, civil society and community partners. It also creates opportunities to align and leverage funding streams. Many existing national and sectoral investments, though not designed specifically for drowning prevention, contribute directly to establishing safer water environments, stronger community preparedness and reduced drowning risks. Recognizing and building on these synergies strengthens both efficiency and impact across government systems.

Fig. 12. A systems-oriented approach for strengthening drowning prevention

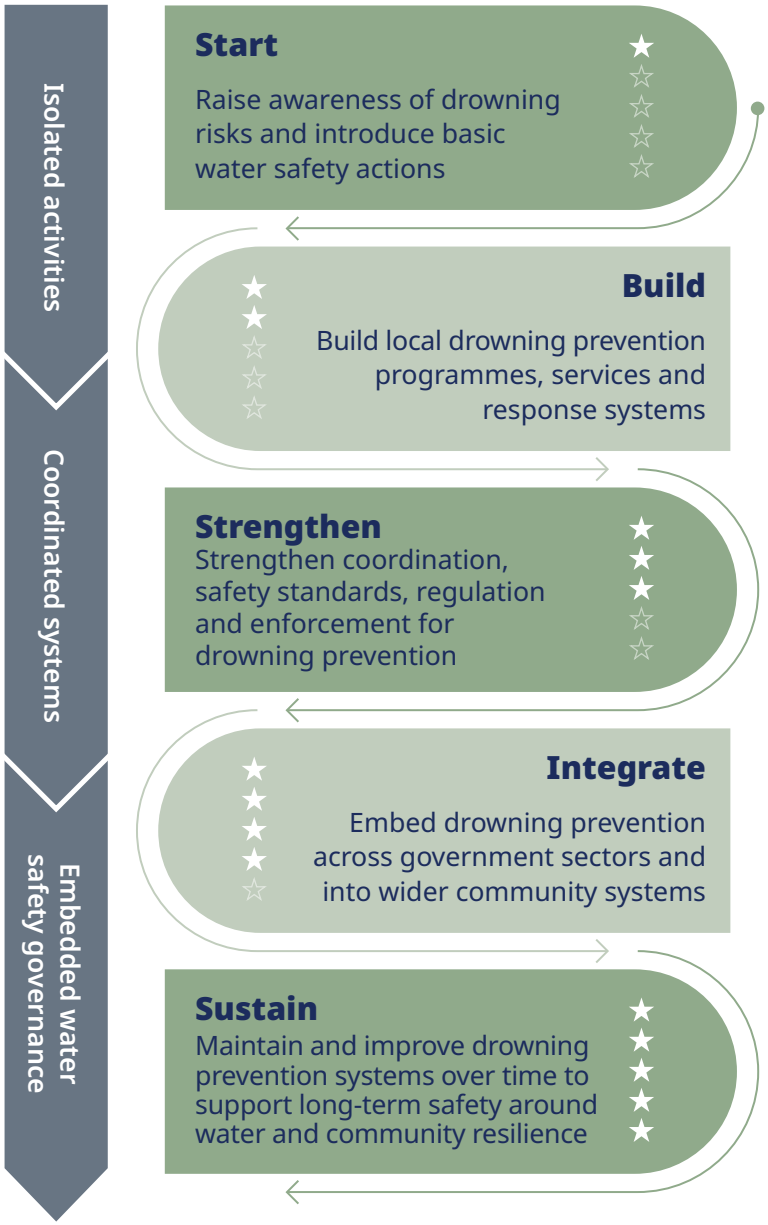
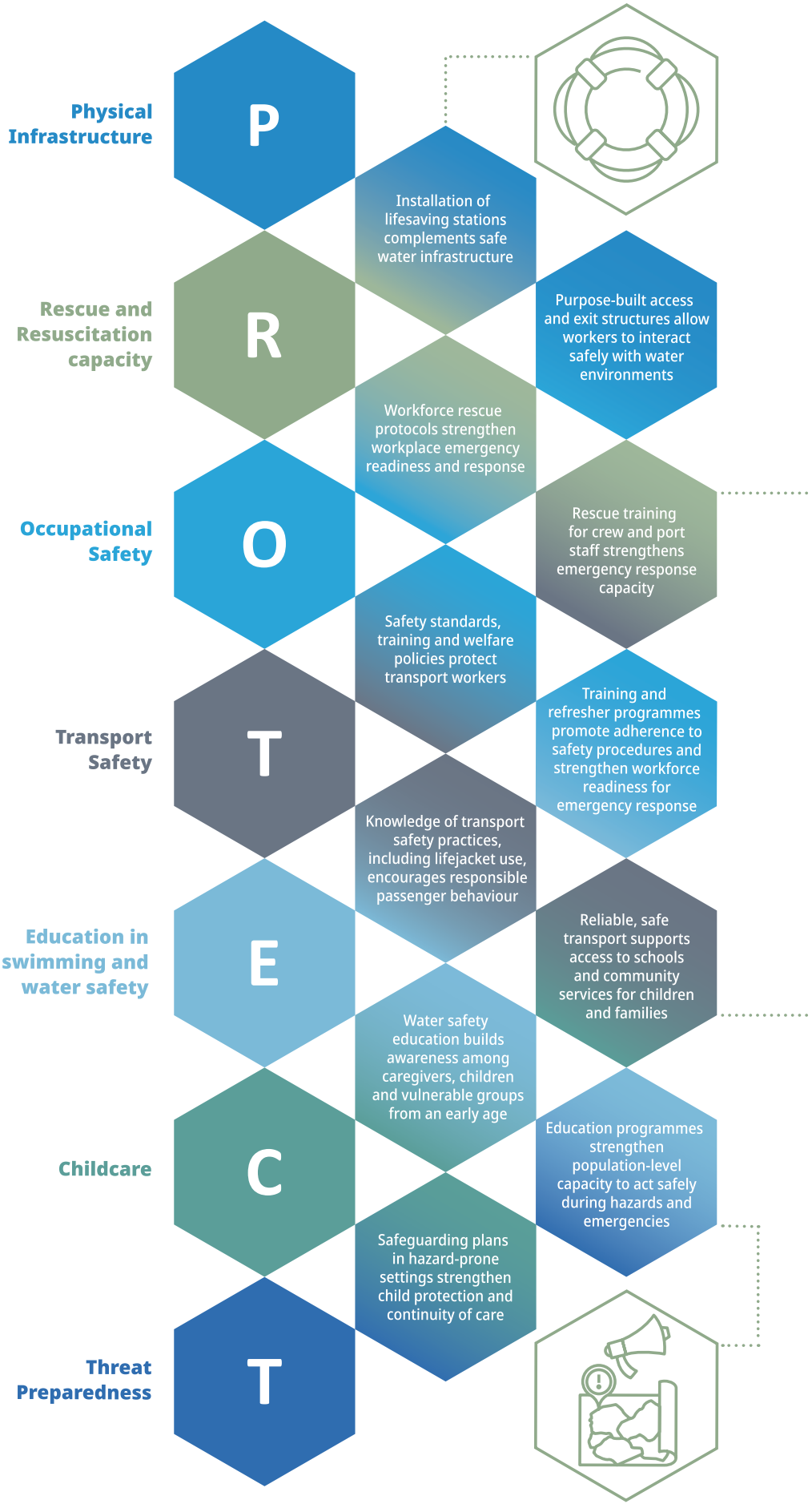


Fig. 13. Multiple co-benefits of each PROTECT recommendation

Read the figure by following the colored intersections between recommendations.

Each intersection represents a co-benefit generated by implementing the connected policy interventions, illustrating how coordinated action across PROTECT strengthens multiple outcomes simultaneously.









Well-designed ports, piers and embarkation areas allow safe boarding and disembarkation

Safe, inclusive learning spaces support swimming and water safety education

CPR and rescue training in education systems build community-wide preparedness

Child-safe barriers and protected play spaces prevent unsupervised child access to water

Trained parents and caregivers can respond safely to water-related emergencies involving children

Nature-based and engineered infrastructure reduces flood risk and strengthens community resilience

Parents trained in water safety through their workplaces pass these skills onto children and communities

Community-based rescue task forces provide timely response during hazards, reducing reliance on professional rescue services

Preparedness measures help maintain business continuity, particularly for small businesses vulnerable to disruptions

Evacuation corridors in the sea and in inland waterways reduce fatalities





Every investment in drowning prevention builds safer communities while delivering broader social, health and economic co-benefits.

Conclusion

The PROTECT technical package builds on existing evidence-based WHO resources on drowning prevention that set out why drowning is not an inevitable consequence of living, working or travelling in proximity to water, but a preventable public health and development challenge that can be reduced through coordinated policy action.

Its seven recommendations provide policy-makers with a practical framework for strengthening safety through protective infrastructure, rescue and resuscitation, occupational settings, transport, education, child care and multi-hazard preparedness. Taken together, policy to promote these actions helps countries move beyond fragmented, standalone efforts by individual sectors towards a coherent, systems-oriented approach that addresses risk at its source, protects those most exposed and embeds safety within the policies and institutions that shape everyday life.

Implementing PROTECT requires political commitment, sustained investment and collaboration across sectors and levels of government, supported by communities, civil society and technical partners. Countries can apply the package in ways that reflect their own priorities, capacities and risk profiles, while building progressively towards stronger governance, better prevention and greater resilience. By investing in drowning prevention, policy-makers can save lives, reduce inequities, strengthen climate and disaster preparedness, and advance broader goals for health, development and social well-being.

The opportunity now is to translate this guidance into action, ensuring that safer interaction with water becomes a shared national responsibility and a lasting policy priority.

Image on next page:
A crew member provides onboard safety support to passengers travelling on a ferry,
© WHO / Jonathon Passmore



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